

## Incentives and the Supply of Effective Charter Schools<sup>†</sup>

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*Charter school funding is typically set by formulas that provide the same amount for students regardless of advantage or need. I present evidence that this policy skews the distribution of students served by charters toward low-cost populations by influencing where charter schools open and whether they survive. To do this, I develop and estimate an equilibrium model of charter school supply and competition to evaluate the effects of funding policies that aim to correct these incentives. The results indicate that a cost-adjusted funding formula would increase the share of disadvantaged students in charter schools with little reduction in aggregate effectiveness. (JEL H75, I21, I22, I28)*

School choice policies aim to improve school quality while expanding educational opportunity. In the United States, charter schools have become the primary vehicle for school choice, with a number of recent papers finding compelling evidence that charter schools are effective at improving student outcomes (Hoxby and Murarka 2009, Abdulkadiroğlu et al. 2011, Dobbie and Fryer 2015, Angrist et al. 2016). These findings have bolstered recent policy momentum behind charter schools, such as the Obama Administration's Race to the Top, based in part on the belief that removing barriers to expansion will lead new, high-performing charter schools to serve underserved student populations.

Two common institutional features of the charter sector call this belief into question, however. First, charter schools, which are publicly funded but privately run, are typically funded by formulas that provide the same amount for all students regardless of advantage or need. Second, although charter schools are unable to screen students, they are differentiated by where they choose to locate and, due to travel costs, serve student compositions that closely reflect local demographics. Taken together, these features raise the question whether the current approach to funding skews the

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distribution of students served by charters toward low-cost student populations by influencing where charter schools decide to open and whether they survive.

To answer this question, I develop and estimate an empirical model of charter school supply and competition. The estimated model allows me to study counterfactual funding policies, in particular a formula that ties revenue to the characteristics of students, in terms of equilibrium sector outcomes. In the model, charter schools simultaneously choose a location in a school district based at least partly on expected revenues, which depend on competition and the statutory per-pupil funding rate, and costs. As variable costs will depend on the composition of students served, the flat formula potentially presents a strategic incentive for new charter schools to spatially “cream skim” and creates differential likelihood of survival of incumbents. Motivated by the evidence on the effectiveness of urban, “No Excuses” charter schools at raising student achievement (Angrist, Pathak, and Walters 2013; Dobbie and Fryer 2013), charter schools are modeled as heterogeneous in their quality, cost structure, and objectives.

This research design requires an institutional setting where the equilibrium location of charter schools reflects their competitive and financial incentives and where those incentives can be measured. In this regard, Florida, which is characterized by a passive regulatory environment and an accordingly high charter penetration rate, is especially well suited. I assemble a unique dataset that links detailed financial records gathered from audits filed by all Florida charter schools with student performance on end-of-grade state exams and school characteristics. I estimate school value-added or effectiveness from the panel of student test scores, while the financial statements of charter schools enable me to separate cost from demand-side determinants of a charter school’s choice of location in the data.

I use the empirical model to evaluate the effects of funding policies on the composition of students served and the aggregate effectiveness of the charter sector. This latter outcome, which is sensitive to competitive incentives, is important for capturing a key policy trade-off: funding policies that raise equity may do so by sustaining ineffective charter schools in the market. Three policy simulations are of interest. First, a cost-adjusted funding formula that ties revenue to student characteristics corrects the financial incentives to skim. I use this counterfactual to answer whether the current funding approach has unintended distortionary consequences. Second, targeted grants for entry into underserved markets may also incentivize charters to serve disadvantaged student populations. Lastly, a general increase in the per-pupil funding rate quantifies the elasticities of charter school supply and effectiveness with respect to funding. The findings thus shed light on the value of expanded social investment in charter schooling.

Estimation of the model presents a number of empirical challenges. In a first step, I estimate the incentive structure of charter school operation, shaped by demand and variable costs, from the data. To do so, I apply a value-added production function to recover school effectiveness, which shifts household demand, from the test score panel. On the supply side, I link effectiveness with charter school expenditures, which I obtain from the financial audits, to estimate variable costs as a function of location and student characteristics. Finally, treating charter school operators, which may be No Excuses, for-profit, or neither, as heterogeneous in their objectives, I leverage revealed preference with exit and location choices in the entry game to

uncover how charter schools respond to competitive and financial incentives and connect those choices to equilibrium aggregate outcomes.

The policy simulations reveal evidence that the flat funding formula leads charter schools to underserve disadvantaged student populations. Implementing a cost-adjusted funding formula generates around a 4 percent increase in the share of subsidized lunch students and an over 7 percent increase in the share of black students attending charter schools. This cost-based cream skimming is driven primarily by non-No Excuses, including for-profit managed, charter schools that the estimates reveal are substantially more cost-oriented in their choice of location. No Excuses charter schools, in contrast, have revealed preferences for serving black and Hispanic student populations. To benchmark the effects on student composition, I compare these results with a policy that would instead regulate where charter schools may open. The policy implemented restricts charter entry only to neighborhoods underserved by public schools. These locations are also disproportionately urban and low income. The results show that the cost-adjusted funding formula closes about 30–50 percent of the discrepancy in charter student demographics with this policy. This result highlights the significant inequities generated by the funding distortion. In addition, because the charter schools sustained in the market by the cost-adjusted policy are not appreciably less effective on average, I find that this gain in equity is associated with little reduction in the aggregate effectiveness of the charter sector.

I find that a targeted start-up grant yields a comparable shift in the location choices of new charter schools to underserved areas as the cost-adjusted formula, but also raises the exit rate of charters that are reallocated by the policy. Furthermore, though the total number of charter schools predictably increases due to an overall increase in charter school funding, aggregate effectiveness responds little. This reinforces the general finding from the counterfactual policy simulations that gains in access to school choice do not appear costly in terms of the quality of charter schooling.

These findings are important as they are informative about school choice policy. In particular, a mismatch between funding and costs may generate significant inequities in access to and benefits from school choice. This point, which is largely unrecognized in both the existing literature on education markets and ongoing policy debate, has potentially broad implications for the design of school choice programs. The findings also underscore that funding policy, an element of both private school voucher and charter school programs, may provide an effective policy instrument for directing competition via supply-side incentives.

The remainder of this paper is organized as follows. In the next section, I situate the paper in the relevant literature. In Section II, I describe the institutional setting of charter schooling in Florida and data sources in detail. I present the empirical model in Section III. Section IV then discusses estimation and identification of the model before I turn to the results, including estimates and counterfactual simulations, in Section V. I conclude in Section VI.

## **I. Related Literature**

School choice reforms embody two policy ambitions. The first is to enhance the quality of public education. Mechanisms supporting this include both direct access to better school alternatives for students and improvements in school quality

stimulated by competition. The second ambition is to expand educational opportunities, especially for underserved students. The growing empirical school choice literature, which combines evidence from international and domestic school choice programs (often private school vouchers and charter schools, respectively), can be viewed as attempts to assess the ability and conditions under which school choice policies may fulfill these ambitions.

A major strand of literature evaluates the effectiveness at improving student outcomes of school alternatives supported by school choice policies. Examples include evaluations of private schools and voucher programs.<sup>1</sup> Evaluations of charter schools typically rely on either student-level administrative data or on admissions lotteries. Papers that estimate charter school effects from changes in exam performance for students who switch between sectors find largely mixed results overall (Sass 2006, Hanushek et al. 2007, Booker et al. 2007).<sup>2</sup> On the other hand, lottery designs provide compelling evidence of improvement in student outcomes (Hoxby and Murarka 2009, Abdulkadiroğlu et al. 2011, Dobbie and Fryer 2015, Angrist et al. 2016).<sup>3</sup> A particular focus of this literature is No Excuses charter schools, which follow an educational philosophy that emphasizes high expectations, comportment, and traditional math and reading skills (Carter 2000, Thernstrom and Thernstrom 2004). No Excuses charters typically serve urban areas and, as findings from New York City and Boston demonstrate (Angrist et al. 2012; Angrist, Pathak, and Walters 2013; Dobbie and Fryer 2013), produce significant positive impacts on student achievement. This evidence has motivated policy efforts to expand charter schooling.

Charter expansion, though, raises questions about the role of competition in education markets. To the degree that households value school quality, school choice creates competitive incentives intended to enhance the quality of public education. This rationale is explicitly written into a number of charter laws, including Florida's. One mechanism for this may be that school choice induces government and public schools to improve. Figlio and Hart (2014), for example, find increases in learning for students who attend public schools disproportionately exposed to private school competition following the introduction of a means-tested voucher program.<sup>4</sup> Papers examining the competitive effects of charters on public school quality, on the other hand, generally produce ambiguous results.<sup>5</sup> Competition may nonetheless discipline the quality of school choice alternatives. Baude et al. (2014) present evidence

<sup>1</sup>International evidence, relying on lottery designs, finds largely positive impacts of attending private schools (Angrist et al. 2002; Angrist, Bettinger, and Kremer 2006; Muralidharan and Sundararaman 2015). In the US context, a number of authors have examined the effectiveness of Catholic private schools (Neal 1997; Altonji, Elder, and Taber 2005). Recent evidence from US voucher programs indicates sharp negative impacts (Abdulkadiroğlu, Pathak, and Walters 2018; Mills and Wolf 2017).

<sup>2</sup>Similarly, CREDO (2009) uses matching techniques with student-level data from 15 states and DC, finding considerable heterogeneity in average charter quality.

<sup>3</sup>Beyond school outcomes, papers using longitudinal and lottery-based approaches have also examined medium and longer term impacts, such as college completion and labor market returns. See Epple, Romano, and Zimmer (2015) for a recent review.

<sup>4</sup>Dinerstein and Smith (2018) study private school responses to increased funding for public schools in New York City, finding that reduced private school supply accounts for a sizable amount of the increase in public school enrollment.

<sup>5</sup>For example, Sass (2006), Booker et al. (2008), and Winters (2012) report positive, if modest, effects, but Bettinger (2005), Bifulco and Ladd (2006), and Zimmer and Buddin (2009) find no competitive effects. Using an IV strategy to overcome endogenous charter location, Imberman (2011) finds mixed or even negative effects.

of improvements in charter school quality in Texas over a ten-year period due to the exit and replacement of low-performing charter schools.

At the same time, scholars have long recognized the potential for school choice policies to generate inequities, compromising the second ambition. While households choosing schools based on characteristics other than school quality will tend to weaken competitive incentives, heterogeneity in preferences may also lead to stratification (Hastings, Kane, and Staiger 2006; Bayer, Ferreira, and McMillan 2007). Weiher and Tedin (2002) present evidence that racial composition predicts households' choice of charter school and Bifulco and Ladd (2007) attribute widening black-white achievement gaps to sorting along racial lines. Such patterns raise the question whether differences across households in the exercise of school choice enables the re-segregation of education.<sup>6</sup> Moreover, how households choose schools and the extent to which preferences for quality vary have supply-side implications that may cut against policy goals. Walters (2018) presents evidence from Boston that although disadvantaged students are most likely to benefit from charter schools, they are significantly less likely to apply and attend.

The role of costs in shaping outcomes remains largely neglected by both of these strands of literature, however. Costs, which influence the supply of school choice alternatives, are important for two reasons. First, fixed costs of entry may undermine competitive incentives by, for instance, leading to excessive or insufficient entry (Spence 1976, Dixit and Stiglitz 1977, Mankiw and Whinston 1986). School choice is only effective if school alternatives are available and underperforming schools face competitive pressure. Second, costs may be an independent source of disparities in education outcomes by generating inequities in access to school choice. With differences in the cost of providing education to disadvantaged students (Duncombe and Yinger 2005), such inequities may stem from distortions in program funding. In recognizing and quantifying the effects of this incentive, this paper thus has parallels with the literature on cost-based cream skimming in health care markets (Ma 1994, Currie and Fahr 2005) and advances the literature on estimation of education costs by linking cost differentials to school behavior and the design of school choice policy (Downes and Pogue 1994).<sup>7</sup>

In this paper, I estimate the full cost structure of charter schools, which includes entry costs and operating cost differences across student populations. The empirical entry model I develop then allows me to connect education market equilibrium outcomes with funding policy through the location and exit choices of charter schools. The estimates also identify differences in objectives across charter schools, including No Excuses charters and those managed by for-profit organizations.<sup>8</sup> In adapting these methods to my setting, the paper fits into the growing empirical literature using these methods to analyze policy questions (Berry and Reiss 2007). Limited prior work has examined the determinants of charter school supply and location, such as

<sup>6</sup>More recently, Ladd, Clotfelter, and Holbein (2017) show that apparent improvements in charter school performance in North Carolina may be driven primarily by positive student selection and Macartney and Singleton (2018) present evidence from North Carolina that suggests charters facilitate "white flight" from public schools.

<sup>7</sup>Relatedly, targeted private school vouchers have been studied as a means of minimizing cream skimming due to peer-based externalities in education production (Nechyba 2000, Epple and Romano 2008).

<sup>8</sup>While the practices and inputs of No Excuses charters have drawn attention in the literature, little work has examined how for-profit charters differ in either behavior or performance. I descriptively examine for-profit management of charter schools in Florida in other work (Singleton 2017).

Glomm, Harris, and Lo (2005) and Hoxby (2006), but these studies take a descriptive approach that is unable to separate demand from cost determinants and does not permit studying counterfactual policy changes. In work related to my approach, Ferreyra and Kosenok (2018) model demand for and entry of charter schools in Washington, DC, finding prospective welfare benefits of expansion, but do not model or consider the implications of strategic behavior by charter schools.<sup>9</sup> This paper differs in focusing on how policies affect the equilibrium quality of charter schooling, which I am able to measure in value-added terms from the panel of student test scores. In addition, the setting I study, in which charter entry has long been relatively unrestricted, is also particularly relevant given policy initiatives aimed at relaxing constraints on charter school expansion.

## II. Background and Data

I describe the institutional background of charter schooling in Florida in this section to highlight the features that make it well suited to studying charter school supply and competition. In particular, Florida provides a setting in which charter schools' entry and location choices reflect their incentives and where those incentives can be measured.

The unique dataset I assemble combines detailed records gathered from financial audits filed by all Florida charter schools with student performance on end-of-grade exams and school characteristics. After describing the data sources for the merged panel, which tracks charter schools between the 2006–2007 and 2012–2013 school years, I also present descriptive findings that motivate the model in this section.

### A. Charter Schooling in Florida

Laws authorizing charter schools aim to strike a balance between operational autonomy and accountability. For instance, charter schools are generally granted considerable independence in terms of curricular design and focus, human resources, and choice of location. To ensure equity, however, charter schools may not screen students or apply any admissions criteria, such as tuition or entrance exams. If a charter school is oversubscribed, an admissions lottery must be held to randomly allocate places in the school. Further, charter schools are not exempt from state accountability and reporting requirements. Florida charter schools, for example, must participate in the School Grades program which assigns letter grades to schools based on student performance on standardized end-of-grade exams.<sup>10</sup> While accountability provisions require intervention in cases of persistently poor performing charters, instances of forced shutdown are relatively rare. A Center for Education Reform report, for example, found that just 18 percent of charter closures nationally were attributable to academic reasons (Consoletti 2011).

Although the level varies by state or district, charter school funding typically provides the same amount for all students served by a school. Florida charter schools, for example, receive per-pupil disbursements through the Florida Education Finance

<sup>9</sup>Also related, Mehta (2017) models competition between charters and public schools in North Carolina.

<sup>10</sup>The School Grades program accounts for both achievement and learning gains.

Program (FEFP) according to a statutory formula. A base funding rate for each full-time equivalent student is multiplied by a district cost differential and totaled. While additional cost adjustments are made for disabled and English language learner students, charter school enrollment of these student populations is low.<sup>11</sup> This funding source may be supplemented by federal programs, such as Title 1, other state sources, or private contributions, but the FEFP program provides the vast majority of operating support for charter schools.<sup>12</sup> The implication of this is that two charter schools serving student populations with different educational needs or advantages, but located in the same school district, will receive largely the same per-pupil support.

Opening of a new charter school must be approved by public school officials. In this regard, Florida's institutional environment is notably favorable to charter growth. For one, unlike other states such as New York or Massachusetts, Florida does not cap the number of charter schools. Moreover, approval to open a new charter school in Florida is based entirely on meeting a set of minimum requirements that includes a curriculum meeting state standards for reading instruction. This review process contrasts sharply from other settings. In New York, for example, applications to open a charter school are also evaluated based on community need. Such need may be registered by excess demand for or lack of existing charter schools, the disadvantages of the target student population, and whether, as measured by proficiency rates of public schools, the community is underserved.<sup>13</sup> Further, while expected success of a charter school is an important factor in other settings, an element that motivates Ferreyra and Kosenok (2018) to model and estimate the regulator's approval problem in Washington, DC, this is also not assessed or considered in Florida. Florida's passive approach instead provides a setting in which the entry and location choices of charter schools can be viewed as reflecting revealed preferences.

The combined effect of Florida's charter-friendly environment is a vital charter sector, characterized by both rapid growth and significant turnover. The first Florida charter schools opened for the 1996–1997 school year. Since then, the sector has expanded into one of the largest nationwide in terms of enrollment and the number of charter schools. For the 2014–2015 school year, 646 charter schools served over 8 percent of all Florida public school students, including more than 13 percent in Broward and Miami-Dade counties.<sup>14</sup> In addition to the high penetration of charter schooling in

<sup>11</sup> The median charter school's student body is around 9 percent disabled and 3 percent English learners in my sample. This compares to 14 percent and 5 percent respectively for the median public school in Florida. Recent evidence indicates this enrollment gap is explained by the combination of a lower baseline probability to apply and a reduced likelihood of special education classification in charter schools (Winters 2015; Winters, Carpenter, and Clayton 2017; Setren 2018).

<sup>12</sup> Charter schools are not considered separate local education agencies in Florida and may receive capital funding or apply for state grants, but these account for little of operational revenues. Florida school districts are not required to share local revenues with charters and almost all do not. However, because Florida's school finance system is largely centralized through FEFP, charters are less disadvantaged in funding relative to public schools than in many other states.

<sup>13</sup> Similarly, Massachusetts requires that at least two charters approved by the state board each year are located in a district in the bottom decile of student performance. A secondary difference between New York and Massachusetts on the one hand and Florida on the other is that state agencies in the former review new charter applications, which specify the intended school district. In Florida, the school districts, which are coterminous with counties, approve charter schools.

<sup>14</sup> As a point of comparison, just 248 charter schools serving about 4 percent of students operated in New York in 2014–2015.

Florida, there is a great deal of entry and exit. For instance, 75 new charter schools opened for the 2013–2014 school year, while 38 shut down and closed.

### B. Data Sources

I combine newly collected financial records for Florida charter schools with existing data on all Florida public schools. Per Florida statute, all charter schools must file an independent financial audit with their district and the state for each year of operation. Containing revenue, itemized expenditures, assets, and capital investment, the audits richly characterize the spending patterns and financial health of each charter school. As a source of financial data, the audits also provide two added advantages (beyond their availability): (i) the audits correspond to an individual school year rather than a fiscal year; and (ii) an audit must be filed for each individual school. I gathered and digitized all audits on file with the Auditor General for the 2006–2007 through 2012–2013 school years.

I use reported total expenditures to measure the variable costs of charter school operation in a given school year.<sup>15</sup> These costs therefore include salaries paid to teachers, staff, and administrators, facility rent or mortgage payments, and any management fee paid, among other expenses incurred during a school year. In using expenditures to measure variable costs, two points of clarification are important. First, these expenditures represent variable costs in that they reflect annually adjustable input purchases and exclude the opportunity costs of entering the market and of operation (which inherently cannot appear on an accounting statement). Second, the expenditures reported in the audits are not necessarily costs at the frontier (i.e., the costs that would be incurred were a charter school efficient in its input usage). Rather, the expenditures embed any allocative inefficiency, presenting a potential challenge for identification.

I merge the expenditure data with indicators for whether each Florida charter school subscribes to a No Excuses philosophy or is for-profit managed. These variables, which I use to estimate heterogeneity across charter schools, are derived from operators' responses to a statutorily-mandated survey by the Florida Office of Independent Education and Parental Choice. To identify No Excuses charters, I combine information regarding mission and instructional specialties with the associations of school philosophy and No Excuses status reported in Angrist, Pathak, and Walters (2013). This is described in greater detail in the Appendix. I identify a charter school as for-profit if its educational philosophy is non-No Excuses and the school reports in the survey that its management organization is for-profit.

Three data sources provide information about the characteristics of Florida charter and public schools. First, I obtain enrollment, grade level, and student body characteristics for all schools from the National Center for Education Statistics' Common Core of Data for the 2006–2007 through 2013–2014 school years.<sup>16</sup> I define a school as open (i.e., operational) if enrollment in any primary grade (K–5) is positive and calculate the number of years since opening for each charter school. Second, I merge these records with the Florida Department of Education's master

<sup>15</sup> I subtract out large capital purchases, such as outlays for buildings or facilities.

<sup>16</sup> For grade-level demographics, subsidized lunch, English language learner, and disabled status, I supplement these data with Florida Comprehensive Assessment Tests (FCAT) Demographic Reports.

school database. This database is used to identify schools as a charter school or not per state records and to obtain each school's exact address.<sup>17</sup> The last data source for school characteristics pertains to their locations and spatial relationship. I geocode the school addresses to 2000 Census tracts, for which I obtain American Community Survey estimates of the local demographics.<sup>18</sup> This is important as it allows me to relate the composition of each school's student body to the composition of the tract where it is located. The spatial distances between tracts also allow me to identify each school's set of competitors.

Finally, I obtain summaries of student achievement by school and grade on end-of-grade Florida Comprehensive Assessment Tests (FCAT) from the Florida Bureau of K-12 Assessment for years 2006–2007 through 2013–2014. Students attending charter and public elementary schools in grades 3 through 5 are examined in math and reading. For each tested grade in a given year, the data contain both the average score and average prior year's score by grade and school for the set of students for whom records for both years are available.<sup>19</sup> Since testing begins in the third grade, I use only fourth- and fifth-grade performance to have a measure of prior learning for all grades. These score averages, which I normalize across schools within grade and year, allow me to estimate the value-added or effectiveness of each school at raising student achievement. I describe the specification and estimation of education production fully later.

My final merged panel tracks 341 charter elementary schools in Florida between the 2006–2007 and 2012–2013 school years.<sup>20</sup> I restrict the sample to non-conversion and non-municipal charter schools, charter schools that are not a virtual school or laboratory school, and charter schools not specialized in serving disabled student populations. Note that 193 (or 57 percent) of the charter schools began operation during the seven-year sample period, while 67 (20 percent) exit at some point. The median panel length is four school years. Included in the sample are also all traditional public elementary schools in Florida.

### *C. Preliminary Evidence*

In this subsection, I present descriptive evidence that motivates key features of the empirical model. I first examine the local market-level determinants in the data of charter school supply in a Census tract before turning to an analysis of charter school exit during the sample.

*Charter School Supply.*—To begin with, I relate the supply of charter schools in a Census tract in the data with characteristics of the local market. For each tract and

<sup>17</sup> A limitation of this data source is that only the present address of charter schools is recorded, so I do not observe moves or model charter school location changes.

<sup>18</sup> I use 5-year Census tract estimates for 2005–2009 through 2009–2013, treating the middle year as corresponding to the spring of that school year. I impute data for 2012 and 2013. I then calculate location characteristics within a given distance of each Census tract using distances between centroids and population weights.

<sup>19</sup> Thus, students not in a Florida public or charter school the prior year are not included in the average. Importantly, the prior scores may have been obtained at another public or charter school if a student switched schools.

<sup>20</sup> I am able to match expenditure data for 92 percent of charter school-year observations. The match rate is particularly low in the school year that a charter school closes, however. The Appendix includes summary statistics for the charter schools in the sample.

year, I calculate the presence of each type of charter school: No Excuses, for-profit, or other. I then stack the panels to estimate regressions of the form

$$(1) \quad \text{Charters}_{jt}^k = \beta^k X_{jt} + \epsilon_{jt}^k.$$

Here,  $\text{Charters}_{jt}^k$  is an indicator variable for whether at least one charter school of type  $k$  is operating in Census tract  $j$  at time  $t$ .<sup>21</sup> The local characteristics in  $X_{jt}$  include public school competition and variables that capture the size of the local market, such as population and household income, which correspond to the area within 1 mile of the tract. To characterize competition, I count the number of public schools within 1 mile, 1–3 miles, and 3–5 miles. I also include the average of reading performance of public schools within 5 miles.

The results, presented in Table 1, show that charter presence is negatively related to local public school performance: the point estimate in column 1, which is statistically significant, suggests that a 1 standard deviation increase in nearby public schools' test scores is associated with around a 0.9 percentage point reduction in the likelihood a charter school is located in the tract. As the baseline presence of charters is around 2 percent, this is a large effect that may stem from both strategic entry and differential exit of charters. Column 1 also suggests that charter presence is positively related to public school presence, but the results in column 2 reveal this association is explained by market size: charter presence is greater in less dense, but more populated areas within school districts (which also tend to have more public schools).<sup>22</sup>

Racial and ethnic demographics of the local market are included as determinants in column 3.<sup>23</sup> These results indicate that charter presence is positively associated with greater shares of black and Hispanic households. All else equal, a 10 percentage point increase in the share of Hispanic households in the area is associated with a 0.004 point (or about 20 percent) increase in the likelihood of charter presence. These associations are suggestive of greater demand for charters among underrepresented minority households, but they also embed any cost differentials that may influence charter school supply on net.

Table 1 is also indicative of important differences across charter schools. In columns 1 through 3, the baseline presence of for-profit managed charter schools in a Census tract is around double the average, while No Excuses charter schools are the least prevalent. This may stem from differences in cost structure, such as greater efficiency of for-profits. Column 4 includes interactions that show the supply of No Excuses schools relative to other types is greater in lower income and higher density markets, suggesting behavioral differences. In addition, though the relationship between the share of underrepresented minorities and supply does not vary across types, column 4 reveals that the negative association with local public school performance is driven primarily by for-profit charters.

<sup>21</sup> Only a handful of tracts have multiple charters of the same type.

<sup>22</sup> Though the estimates are not presented in Table 1, the associations with public school presence in the 1–3 and 3–5 mile distances are insignificant throughout.

<sup>23</sup> These characteristics of households and students proxy for differences in socioeconomic status, household wealth, and other sources of advantage.

TABLE 1—DETERMINANTS OF CHARTER SCHOOL PRESENCE

|                                         | (1)                   | (2)                   | (3)                    | (4)                     |
|-----------------------------------------|-----------------------|-----------------------|------------------------|-------------------------|
| Public $\bar{z}$                        | −0.00943<br>(0.00419) | −0.00893<br>(0.00403) | −0.00693<br>(0.00416)  | −0.000185<br>(0.00448)  |
| No Excuses $\times$ public $\bar{z}$    |                       |                       |                        | −0.00403<br>(0.00429)   |
| For-profit $\times$ public $\bar{z}$    |                       |                       |                        | −0.0162<br>(0.00575)    |
| Number of publics < 1 mile              | 0.00399<br>(0.00196)  | 0.00231<br>(0.00205)  | 0.00173<br>(0.00182)   | 0.00173<br>(0.00182)    |
| log population                          |                       | 0.00618<br>(0.00228)  | 0.00407<br>(0.00218)   | 0.00407<br>(0.00218)    |
| log density                             |                       | −0.00655<br>(0.00154) | −0.00656<br>(0.00162)  | −0.00341<br>(0.00221)   |
| No Excuses $\times$ log density         |                       |                       |                        | 0.000453<br>(0.00156)   |
| For-profit $\times$ log density         |                       |                       |                        | −0.00989<br>(0.00350)   |
| log HH income                           |                       | −0.00298<br>(0.00316) | 0.00117<br>(0.00226)   | 0.00481<br>(0.00261)    |
| No Excuses $\times$ log HH income       |                       |                       |                        | −0.00596<br>(0.00344)   |
| For-profit $\times$ log HH income       |                       |                       |                        | −0.00498<br>(0.00625)   |
| Percent black                           |                       |                       | 0.000282<br>(0.000114) | 0.000366<br>(0.000173)  |
| Percent Hispanic                        |                       |                       | 0.000440<br>(0.000148) | 0.000524<br>(0.000225)  |
| No Excuses $\times$ percent UR minority |                       |                       |                        | −0.000183<br>(0.000134) |
| For-profit $\times$ percent UR minority |                       |                       |                        | −7.13e-05<br>(0.000272) |
| No Excuses                              | −0.0102<br>(0.00368)  | −0.0102<br>(0.00368)  | −0.0102<br>(0.00368)   | 0.0587<br>(0.0377)      |
| For-profit                              | 0.0227<br>(0.00846)   | 0.0227<br>(0.00846)   | 0.0227<br>(0.00846)    | 0.152<br>(0.0778)       |
| $R^2$                                   | 0.019                 | 0.021                 | 0.023                  | 0.025                   |

Notes: Standard errors clustered by school district in parentheses. 66,213 Census tract-by-charter school type-year observations. All characteristics correspond to area within one mile of the Census tract. All columns include school district and year fixed effects. Specifications also control for the number of public schools within 1–3 and 3–5 miles, and the shares of Asian and other demographic households in columns 3 and 4. *Percent UR minority* is sum of percent black, percent Hispanic, and percent other.

Though these determinants of charter supply are informative, they remain limited in significant respects. For one, the market-level correlates don't speak to competition between charter schools. Further, the associations, particularly with household demographics, are unable to separate the influences of demand versus cost-side factors. The results are thus ambiguous about the role of a funding formula distortion in structuring market outcomes. To shed light on these mechanisms, I turn next to an analysis of the exit margin.

*Charter School Exit.*—For motivation, I present summary statistics in Table 2 that compare charter schools that exit the sample with those that persist (i.e., are

TABLE 2—CHARACTERISTICS OF EXITERS AND PERSISTERS, 2006–2013

|                                 | Exiters | Persisters | Difference |
|---------------------------------|---------|------------|------------|
| <i>Location characteristics</i> |         |            |            |
| Density                         | 4,468   | 4,039      | 429        |
| Household income                | 46,366  | 62,713     | –16,347    |
| Percent white                   | 37.46   | 47.56      | –10.10     |
| Percent black                   | 24.87   | 12.18      | 12.69      |
| Percent Hispanic                | 34.06   | 36.54      | –2.47      |
| Percent Asian                   | 1.65    | 2.11       | –0.46      |
| <i>Student characteristics</i>  |         |            |            |
| Enrollment                      | 287.71  | 408.78     | –121.10    |
| Percent FRP lunch               | 66.23   | 45.27      | 20.97      |
| Percent white                   | 15.93   | 39.00      | –23.07     |
| Percent black                   | 53.86   | 17.08      | 36.78      |
| Percent Hispanic                | 28.52   | 39.93      | –11.40     |
| Percent Asian                   | 1.43    | 2.75       | –1.32      |
| <i>School characteristics</i>   |         |            |            |
| Expenditure/pupil               | 7,988   | 7,086      | 902        |
| Math z                          | –0.90   | 0.05       | –0.96      |
| Reading z                       | –0.62   | 0.17       | –0.79      |
| No Excuses                      | 0.06    | 0.18       | –0.12      |
| For-profit                      | 0.12    | 0.37       | –0.25      |

*Notes:* Location characteristics represent area within one mile of a school's Census tract. Math and reading scores represent averages of within grade and year normalized fourth and fifth grade exam performance.

either open in 2007–2008 or after, but do not exit by 2012–2013).<sup>24</sup> The comparisons reveal a number of striking differences between exiter and persister charter schools. In particular, the locations of charter schools that exit are higher density and much lower income on average (about \$16,000 less mean household income). Accordingly, the average exiter charter school serves a 21 percentage point larger share subsidized lunch student composition. Exiters also serve a 37 percentage point larger share black student composition on average.

The bottom panel of Table 2 compares exiter and persister charter schools in terms of expenditure per pupil, type, and student performance. Charter schools that exit spend \$900 more per pupil (approximately 10 percent of total expenditure) per year than those that persist in the sample. Exiters are also less likely to be either No Excuses or for-profit managed. In terms of math and reading performance, students attending exiter charter schools perform dramatically worse on average. The average math score for exiter charter schools, for example, is nearly 1 standard deviation on average below that of persister charter schools. The interpretation of these differences is nevertheless unclear: though exiters both serve more disadvantaged student populations and spend more per pupil on average, this may stem from where they locate rather than cost differences across student populations. Similarly, the large differences in student performance may indicate differences in underlying quality, consistent with market efficiency.

<sup>24</sup>To do this, I use the full panel to compute conditional averages for exiter and persister charter schools and test for differences by regressing each variable on indicators for exiter, persister, and entrant, age, dummies for the year a school enters or exits, and year fixed effects.

To better understand these patterns, I examine how characteristics of a charter school predict exit. To do this, I estimate linear probability models of the form

$$(2) \quad \text{Exit}_{it} = \beta f(\mathbf{Z}_{it}) + \gamma X_{it} + \varepsilon_{it},$$

where  $\text{Exit}_{it}$  is an indicator variable for whether charter school  $i$  exits the sample between period  $t$  and  $t + 1$ . The vector  $\mathbf{Z}_{it}$  represents the characteristics of the student body at charter school  $j$ , including demographics and free or reduced-price lunch status. I use  $f(\cdot)$  to indicate that I allow for nonlinearities (interactions and quadratic terms) in the share of each student group in the specification. The term  $X_{it}$  includes the presence of public and competing charter schools nearby, total enrollment, student performance in reading, age, and characteristics of the school's Census tract, including density and demographics.

Column 1 of Table 3 relates charter school exit to local competition, characteristics of the student body, and the age of the school. While the market-level determinants indicate charter presence accompanies low public school performance, these results suggest a limited influence on the exit margin. Moreover, greater competition in the form of either more public or charter schools nearby does not appear to influence survival.<sup>25</sup> Student demographics, on the other hand, are associated with exit: the estimates in (1) indicate that a 10 percentage point increase in the share of black students attending a charter school is associated with a 0.02 point increase in the probability of exit, all else held constant. This magnitude represents an over 40 percent increase in the exit probability from the baseline level.

Column 2 conditions on enrollment itself rather than competition while column 3 also controls for student exam performance in reading, which differs dramatically between exiters and persisters. The results indicate that larger and higher-performing charter schools are less likely to exit, though the latter effect is not precisely estimated. The effect of enrollment suggests that, to the degree that competition draws away students, it will raise the probability of exit. At the same time, the share of black students remains positively associated with exit and similar in magnitude across specifications. To absorb unobserved type-specific differences across schools, column 4 also controls for charter school type. While the association with the share of black students is attenuated, student demographics continue to predict exit above and beyond location characteristics, charter school size, and student exam performance.

These results can be viewed as consistent with cost differences across students that are not reflected in the funding formula: charter schools' enrollment (in combination with the flat funding formula) implicitly conditions the comparisons on expected revenue. The independent variation in enrolled student composition then proxies for cost differentials, which predict survival even for otherwise observably similar charter schools. As indirect evidence that funding formula incentives are distortionary, this interpretation must be applied with caution. Incumbent charter schools are a selected sample and the results are not able to rule out unobserved differences that may influence survival.

<sup>25</sup>The effects of charter and public presence at greater distances, while not presented in Table 3, are also not significantly related to exit.

TABLE 3—MARGINAL EFFECTS ON PROBABILITY OF CHARTER SCHOOL EXIT

|                             | (1)               | (2)               | (3)               | (4)               |
|-----------------------------|-------------------|-------------------|-------------------|-------------------|
| Public $\bar{z}$            | -0.012<br>(0.015) |                   |                   |                   |
| Number of publics < 1 mile  | -0.014<br>(0.010) |                   |                   |                   |
| Number of charters < 1 mile | -0.011<br>(0.008) |                   |                   |                   |
| Percent FRP lunch           | 0.001<br>(0.000)  | 0.001<br>(0.000)  | 0.001<br>(0.000)  | 0.001<br>(0.000)  |
| Percent black               | 0.002<br>(0.001)  | 0.002<br>(0.001)  | 0.002<br>(0.001)  | 0.001<br>(0.001)  |
| Percent Hispanic            | -0.000<br>(0.001) | 0.000<br>(0.001)  | -0.000<br>(0.001) | -0.000<br>(0.001) |
| Percent Asian               | -0.009<br>(0.005) | -0.008<br>(0.005) | -0.007<br>(0.005) | -0.009<br>(0.005) |
| log age + 1                 | -0.033<br>(0.013) | -0.026<br>(0.012) | -0.027<br>(0.012) | -0.033<br>(0.013) |
| log enrollment              |                   | -0.027<br>(0.011) | -0.028<br>(0.012) | -0.028<br>(0.012) |
| Reading $z$                 |                   |                   | -0.005<br>(0.011) | -0.003<br>(0.011) |
| No Excuses                  |                   |                   |                   | -0.067<br>(0.017) |
| For-profit                  |                   |                   |                   | -0.074<br>(0.021) |
| Baseline $\Pr(\text{Exit})$ |                   | 0.046             |                   |                   |
| $R^2$                       | 0.183             | 0.181             | 0.197             | 0.212             |

Notes: Standard errors clustered by school in parentheses. 1,451 charter school-year observations. All columns include school district and year fixed effects. Specifications also control for the number of charters and publics within 1–3 and 3–5 miles each in columns 1 and 2, household demographics within 1 mile of the school's Census tract, the percentages of ELL, ESE, and gifted students, share of enrollment in grades > 5, an indicator for year of entry, and Census tract income quintile-by-urban intercepts.

Informed by these data patterns, the empirical model, which I describe in the next section, aims to directly quantify the mechanisms that underlie equilibrium charter school supply by estimating the cost structure and objectives of charter schools.

### III. Empirical Model

This section presents the empirical model of charter school supply and competition that I develop. The model links the competitive and financial incentives of charter schools with their location and operation choices. For this purpose, there are two key elements of the model. First, I allow charter schools to differ in their effectiveness at raising student achievement (i.e., value-added), which is valued by households on the demand side. This allows for competition to drive ineffective charter schools from business and for assessing the equilibrium effects of funding policy on the quality of the charter sector. Second, the variable costs of operating a charter school depend on the characteristics of students served in the model. Given the statutory funding formula, charter schools may therefore strategically locate and incumbents' survival may be linked to the students they serve.

The basic building block of the model is an elementary education market, which I consider to be a school year and district combination. In each market, the following sequence occurs. First, incumbent charter schools, those in operation during the prior school year, decide whether to exit. Simultaneously, a pool of potential entrant charter schools decide whether to enter the market. These choices define the set of active schools from which households may choose from. Households then choose a school to attend at the beginning of the school year and student achievement is realized at the conclusion of the school year.

Charter schools' entry and exit choices are made in an incomplete information setting. Given the payoff-relevant information summarized by a commonly observed state vector, their choices constitute a Bayesian Nash equilibrium. The state vector then evolves between periods as a result of those choices. I adapt this basic setup to my setting in a couple ways. First, competition contains a spatial dimension and entering charter schools choose a location, characterized as a bundle that includes expected revenues (that depend on the per-pupil funding rate) and costs. Second, in addition to differing in their effectiveness, charter schools are heterogeneous in their objectives in the model. I allow for three types of charter schools: No Excuses, for-profit managed, and a nonprofit but non-No Excuses type.

While the model endogenizes the composition of students served by charter schools and the aggregate quality of the charter sector, I abstract from other aspects of the education market. For instance, though they compete for students, the locations and effectivenesses of public schools are treated as exogenous.<sup>26</sup> Additionally, I do not model changes in effectiveness over time at the school level, though entry and exit dynamics may shift effectiveness in the aggregate. Instead, effectiveness is an idiosyncratic characteristic drawn post-entry, and thereby a source of uncertainty in the model, that evolves in a deterministic fashion.

This section is divided into three parts. In the first subsection, I define a school's effectiveness as its contribution to student learning in the context of a value-added education production function. In the second subsection, I detail the elements of the location choices of potential entrant charter schools and exit choices of incumbent charter schools, which are predicated on the "primitives" of demand and costs. I describe the equilibrium concept and evolution of states in the last subsection. Throughout, I suppress district-specific notation to simplify the exposition.

### *A. Education Production*

In the model, both public and charter schools combine their inputs with those of households to produce education outcomes, measured by student performance on end-of-grade exams. Performance also depends on accumulated impacts of inputs up to the present, which I assume are fully summarized by a function of exam performance the prior year. As a result, education is produced in a value-added fashion:

$$(3) \quad A_{nit}^k = \rho^k(A_{nt-1}^k) + \beta^k Z_{nt} + \mu_{it}.$$

<sup>26</sup>There is weak to mixed evidence for competitive effects of charter schools on public school students (Epple, Romano, and Zimmer 2015).

In this equation,  $A_{nit}$  is student  $n$ 's achievement from attending school  $i$  during school year  $t$ , while  $A_{nt-1}$  represents prior learning. The  $k$  superscripts index subjects, e.g., math and reading. The contribution of contemporaneous household inputs to learning is represented by  $\beta^k Z_{nt}$ .

I term the contribution or value-added of school  $i$  to student learning gains, represented by  $\mu_{it}$ , its effectiveness. This term summarizes the productive contribution of all of a school's inputs to student learning, including teacher quality, learning environment, infrastructure, management, etc., in a single index. In this way, effectiveness, which might equally be thought of as quality, vertically differentiates schools in the model.

### B. Charter School Utility

Charter schools make entry and exit choices to maximize expected utility in the model.<sup>27</sup> While these choices may depend on expected revenues and costs, I allow for heterogeneous objectives across charter schools by type.

The latent utility for charter school  $i$  operating in location  $j$  at state  $s_t$  is given by

$$(4) \quad U_{ij}(s_t) = U_i(\Pi_{ij}(s_t), \mathbf{D}_{ij}(s_t), \mu_{it}),$$

where  $\Pi_{ij}(s_t)$  is net income and  $\mathbf{D}_{ij}(s_t)$  is a vector of student enrollments in  $i$  in location  $j$ . This formulation of utility, which shares commonalities with numerous health sector applications (Lakdawalla and Philipson 2006), is intended to capture charter school operation for potentially not-for-profit, altruistic, or mission-oriented reasons (Malani, Philipson, and David 2003). An important implication of this utility is that the degree to which charter schools respond to financial incentives is an empirical question. I model heterogeneity in charter school objectives by allowing the utility function to vary with type in the estimation.

Represented by  $\Pi_{ij}(s_t)$ , net income is determined by revenue, variable costs, and commonly known fixed costs of operation,  $FC_{jt}$ :

$$(5) \quad \Pi_{ij}(s_t) = \tau_t D_{ij}(s_t) - VC_{ij}(s_t) - FC_{jt}.$$

The product of the funding rate,  $\tau_t$ , and total enrollment,  $D_{ij}(s_t)$ , yields revenue. Each location in a school district,  $j$ , is associated with an enrollment level and variable costs of operation,  $VC_{ij}(s_t)$ . Charter schools' choices of where to locate or whether to exit therefore depend on these "primitives." I describe the structure I place on these two objects, from which competitive and strategic incentives derive, in turn next.

*Demand.*—Households' school choice determines charter school enrollment in the model. Households weigh characteristics, including effectiveness, which may be imperfectly observed, and travel distance in selecting a school to attend from among the available alternatives in their district. As a result, the locations and

<sup>27</sup> While, as detailed in Section IIIC, charters take expectations over other charter schools' simultaneous choices, they are not forward-looking in the model. I discuss potential biases from this restriction in the estimation results.

effectiveness of public and other charter schools have competitive implications for charter schools' enrollment.

For parsimony, I abstract from the household-level choice problem to model the enrollment of a given household type  $z$  at each charter school as a function of its effectiveness, location characteristics, and competition. Household types include demographic and socioeconomic groups. The total enrollment of student group  $z$  in school  $i$  in location  $j$  is given by

$$(6) \quad D_{ij}^z(s_t) = D_i^z(\mu_{it}, x_{jt}^D, a_{-it}, \mu_{-it}).$$

In this expression, demand depends in part on the effectiveness,  $\mu_{it}$ , of a charter school as well as the characteristics of the local market  $j$ , which are included in  $x_{jt}^D$ . These local market characteristics capture the size of the market for household type  $z$ . Reflecting competition,  $a_{-it}$  and  $\mu_{-it}$  represent the locations and effectivenesses, respectively, of competing charter and public schools.

This formulation of demand captures a number of features of education markets. First, the demand function incorporates competitive effects of nearby public and competing charter schools, which may be magnified by closer proximity. In this way, competition between schools is spatial, reflecting the imperfectly competitive nature of the education market. Second, effectiveness, to the degree that it is valued by households, also differentiates schools in that, all things being equal, greater own effectiveness increases enrollment. These features are important for modeling the horizontal and vertical aspects of school competition. Finally, the formulation allows for heterogeneity in how households, characterized by type, evaluate schooling alternatives.

From (6), the total enrollment for a charter school  $i$  in location  $j$  is given by summing demand over mutually exclusive household types,

$$(7) \quad D_{ij}(s_t) = \sum_z D_{ij}^z(s_t).$$

Similarly, the vector of household type shares of students attending each charter school, which I denote by  $\mathbf{Z}_{ij}(s_t)$ , is endogenized in the model by (6) and (7).

*Variable Costs.*—Operating a charter school incurs costs that can be conceptually separated into variable and fixed. Variable costs represent the minimum expenditure for a charter school in a given state. Unlike fixed costs, variable costs therefore depend on input prices, which may vary across locations and with student composition, and a charter school's outputs. These relationships are summarized by charter schools' variable cost function,

$$(8) \quad VC_{ij}(s_t) = VC_i(D_{ij}(s_t), \mu_{it}, \mathbf{Z}_{ij}(s_t), x_{jt}^C).$$

Expressed in this way, each charter school is a "firm" that produces effectiveness,  $\mu_{it}$ , for a given enrollment,  $D_{ij}(s_t)$ , and composition of students,  $\mathbf{Z}_{ij}(s_t)$ . As effectiveness represents a combination of inputs, this formulation recognizes it as potentially costly to supply.

Since student composition is tied to a charter school's location through demand, a strategic incentive to "cream skim" potentially derives from the dependence of costs on the student composition served in equation (8). While I do not model the mechanism through which they arise, such cost differentials may in general stem from two sources: the production surface or input prices. For instance, teaching assistants or non-classroom school inputs, such as social services, may be especially important for effectively serving disadvantaged students. This is consistent with recent evidence from school finance reforms that spending improves later life outcomes of low socioeconomic status students in part by increasing support services and staff (Jackson, Johnson, and Persico 2016).<sup>28</sup> As a result, the optimal mix of inputs may vary with the composition of students. To input prices, evidence from teacher sorting suggests that teachers view concentrations of disadvantaged students as a disamenity, necessitating compensating differentials (Lankford, Loeb, and Wyckoff 2002; Jackson 2009; Clotfelter, Ladd, and Vigdor 2011).<sup>29</sup> I assume that input prices are fully captured by the combination of a charter school's student composition and location-specific characteristics,  $x_{jt}^C$ .

### C. Entry Location and Exit

Based on their status in the prior period, charter schools in the model are either incumbents, who choose whether to exit, or potential entrants, who choose whether to enter and a location. These choices are made simultaneously in an incomplete information setting in which the actions of competitor schools are stochastic from the point of view of any given charter school. This structure arises from privately known preference draws.<sup>30</sup> In addition, the effectiveness of potential entrants is only revealed (to them and to rivals) post-entry in the model.

For describing charter schools' choice problems, it is useful to divide the state vector,  $s_t$ , into four subcomponents. First,  $a_{-it}$  represents the actions of other charter schools, which may be either a location in the market or decision to exit/remain out ( $a_{it} = 0$ ), and the locations of public schools. Second,  $s_{it}$  lists the characteristics of charter school  $i$ , which include its effectiveness, its type, its age and incumbency (i.e., whether in operation the prior period). Vector  $s_{-it}$ , on the other hand, lists the characteristics of charter schools other than  $i$  and of public schools. The variable  $x_t$  summarizes the exogenous characteristics of all market locations that shift demand and costs (including the per-pupil funding rate,  $\tau_t$ ).

Incumbent charter schools make the choice of remaining in operation in their present location,  $j$ , or exiting the market. Expected utility of operating is given by  $E[U_{ij}(s_t)] = E[U_j(a_{-it}, s_{it}, s_{-it}, x_t)] + \epsilon_{ijt}$ , where  $\epsilon_{ijt}$  is an idiosyncratic draw. The expectation is taken over competing charter schools' simultaneous entry and exit decisions and the effectivenesses of potential entrants. I normalize the utility of

<sup>28</sup> Relatedly, students in schools with concentrated low-income populations disproportionately benefit from non-instructional spending (Sorensen 2016).

<sup>29</sup> Note that charter schools are not constrained by teacher salary schedules. Evidence indicates that hiring practices in charter schools are more market-based than traditional public schools (Podgursky 2006). Differences across households in terms of parental involvement in the school may also translate to cost differences.

<sup>30</sup> Though I refer to them as "preference draws" because they are additive with utility, these choice shocks may reflect private information about demand or costs or other conditions that affect the utility of a given choice alternative.

exiting to 0. Thus, incumbent  $i$ 's choice problem can be expressed as to remain in operation (in location  $j$ ) if

$$(9) \quad E[U_j(a_{-it}, s_{it}, s_{-it}, x_t)] + \epsilon_{ijt} \geq \epsilon_{i0t},$$

and to close otherwise.<sup>31</sup> The terms  $\epsilon_{ijt}$  and  $\epsilon_{i0t}$  are  $i$ 's private information regarding the utility of continuing operation and exiting, respectively.

For a charter school deciding to enter the market, the choice problem also contains a location choice. An entering charter school weighs the alternative locations in a school district, indexed by  $j$ , by computing the expected utility of each alternative. Entering the market incurs known entry costs that are denoted  $EC_t$ , which represent both monetary and nonmonetary costs associated with the application, authorization, and opening process. The choice problem facing potential entrants can be written as

$$(10) \quad \max \left\{ E[U_j(a_{-it}, s_{it}, s_{-it}, x_t)] - EC_t + \epsilon_{ijt}, \dots, \right. \\ \left. E[U_J(a_{-it}, s_{it}, s_{-it}, x_t)] - EC_t + \epsilon_{iJt}, \epsilon_{i0t} \right\}.$$

In addition to the discrete set of locations in the school district, potential entrants may choose to remain out of the market, yielding only the associated private information draw,  $\epsilon_{i0t}$ .

#### D. Equilibrium and State Evolution

Equations (9) and (10) map charter schools' expected utility, given their beliefs about the choices of other charter schools, into ex ante choice probabilities. Since the choice shocks are private information, each charter school's decision rule is a function of the common state variables and its private information, but not the private information of other charter schools. I denote by  $\mathbf{P}_t$  the set of all charter schools' choice probabilities to re-express the expected utility for charter school  $i$  of location  $j$  as

$$(11) \quad E[U_j(a_{-it}, s_{it}, s_{-it}, x_t)] = \tilde{u}_{ij}(s_t, \mathbf{P}_t).$$

Given this expected utility, (9) and (10) represent a system of best response probability functions that define the Bayesian Nash equilibrium of the game wherein strategies and beliefs are consistent.<sup>32</sup> Charter schools' choices determine the equilibrium outcomes in the current period and the state vector evolves as a result of those choices.

<sup>31</sup> In treating exit from the market in this way, the model abstracts from dynamics such as "unraveling" or "tipping" that might lead charter schools to exit. In the Appendix, I present evidence that these dynamics are of limited importance. Table A2 shows that within-school changes in enrollment, student demographics, and test scores are not significantly related with years until a charter school exits in the data.

<sup>32</sup> I focus on pure-strategy equilibria, with anonymous and symmetric strategies. Existence of equilibrium follows from Brouwer's fixed point theorem, but is not guaranteed to be unique.

The vector of charter schools' effectivenesses,  $\mu_t$ , is a state that influences choices and thereby outcomes. While competitive entry and exit dynamics endogenize the aggregate effectiveness of the charter sector in the model, I specify a deterministic law of motion for the evolution of a given school's effectiveness between periods:

$$(12) \quad \mu_{it} = \mu_{1i} + \mu_2 \log(e_{it} + 1).$$

In this process,  $\mu_{1i}$  represents an idiosyncratic and permanent component of effectiveness that is drawn upon entry, while  $\mu_2$  scales the rate at which effectiveness accumulates with age,  $e_{it}$ , of a charter school. This process, which is similar to returns to teacher quality with experience, captures improvement over time within charter schools.

#### IV. Estimation and Identification

Estimation of the empirical model consists of three steps: (i) recovering effectiveness of schools from the panel of test score data; (ii) estimating the demand and variable cost functions, which shape the incentive structure of charter school operation; and (3) plugging in the "offline" estimates to estimate how charter schools respond to the incentive structure via revealed preference. In this section, I detail each estimation step and discuss the identification of the model parameters.

##### A. Offline Elements

*Effectiveness.*—I use the panel of end-of-grade exam scores by school and grade to estimate the effectiveness of each school at raising student achievement. Averaging (3) to the grade level and substituting in charter schools' law of motion for effectiveness (12) yields the equation that I take to the data:

$$(13) \quad A_{igt}^k = \rho_g^k(A_{igt-1}^k) + \beta^k \mathbf{Z}_{igt} + \mu_{1i} + \mu_2 \log(e_{it} + 1) + \delta_{gt}^k + \nu_{igt}^k.$$

The dependent variable,  $A_{igt}^k$ , is the normalized average subject  $k$  score of grade  $g$  students in school  $j$  at time  $t$ , while  $A_{igt-1}^k$  is their average score the prior year.<sup>33</sup> Net of subject-grade-year intercepts,  $\delta_{gt}^k$ , effectiveness at raising student achievement is constant across subjects and grades.

The primary identification challenge in recovering estimates of school effectiveness stems from potential household sorting on unobservables across schools, which would lead to correlation between effectiveness and the error term,  $\nu_{igt}^k$ . As a result, the control for prior year's achievement in (13), assumed to capture unobserved inputs into student learning prior to  $t$ , is key to guard against this. Recent evidence regarding teachers' value-added, for example, indicates that controlling for prior student test scores is most important for obtaining unbiased causal estimates (Chetty, Friedman, and Rockoff 2014). I control for grade- and subject-specific cubic

<sup>33</sup> Note that the averaging in both years is over the same students, although the prior score may have been received at a different Florida public or charter school if the student switched schools. I normalize current and prior scores separately.

polynomials in the prior score. In addition, I control for all observed grade-level student characteristics, including demographics, subsidized lunch, special education, English language learner, and gifted student status in  $\mathbf{Z}_{igt}$ .<sup>34</sup> I also interact the student characteristics with a charter indicator to allow for differential sorting across the public and charter sectors.

I estimate (13) by school fixed effects, pooling charter and public schools. I weight each observation in the data by the number of students entering into the calculation of the achievement score averages in the estimation. As raw fixed effect estimates typically overstate the variance of school value-added (Kane and Staiger 2008; Deming 2014; Koedel, Mihaly, and Rockoff 2015), I use empirical Bayes techniques to shrink the estimated fixed effects. The adjusted estimates for charter schools are taken as estimates of  $\mu_{1i}$ , while the effectivenesses of public schools are assumed to be fixed over time. Residual average growth in student performance across grades and years therefore identifies the effectiveness of each school.

*Demand.*—To estimate charter school demand, I pool the elementary enrollment of household types across charter schools and years. I use Asian, black, Hispanic, white, other demographic, and subsidized lunch status, a proxy for household income, as the types of households. To capture spatial competition, I then bin counts of nearby charter and public schools into mutually exclusive distance bands centered around each school's location, while conditioning on charter school effectiveness and location characteristics.

The equation I estimate is given by

$$(14) \quad \log(D_{it}^z + 1) = \alpha^z(\mu_{it} - \bar{\mu}_{j(i)t}^P) + \sum_d \gamma_c^d NC_{j(i)t}^d + \sum_d \gamma_p^d NP_{j(i)t}^d + \varrho^z x_{j(i)t}^D + \xi_{it}^z,$$

where  $NC_{j(i)t}^d$  and  $NP_{j(i)t}^d$  represent the number of charter and public schools, respectively, in distance band  $d$  from charter school  $i$ 's location at time  $t$ . The  $\gamma$  parameters are thus interpretable as semi-elasticities of enrollment with respect to an additional competitor (public or charter) within a given distance band. Like the preliminary analysis, I use three distance bands in the estimation: within one mile, between one and three miles, and between three and five miles.

To capture the local market size, I similarly include the logged school aged population of each demographic group within the one-, three-, and five-mile distance bands of the Census tract in  $x_{j(i)t}^D$ . In addition to controlling for the population of the “own” (type  $z$ ) demographic nearby (e.g., the enrollment of Hispanic students depends on the population of Hispanics in the local market), the specification thus allows for “cross” spillovers of other populations on enrollment. This dependence is consistent with the evidence on school sorting that indicates preferences for peer groups influence households' school choice (Rothstein 2006). Note that such preferences over peers may be direct or induced by preferences over test score levels (which households may use to infer effectiveness).

<sup>34</sup> Since I do not observe the characteristics of just the matched sample of students (those for whom current and prior scores are available), I assume that the overall grade-level demographics are representative. I also control for the share of grade-level enrollment that the matched test score sample accounts for in the estimation.

Demand also depends on charter school  $i$ 's effectiveness relative to the quality of nearby public schools, given by  $\mu_{it} - \bar{\mu}_{j(i)t}^P$  in equation (14). For the latter term, I compute the average effectiveness of public schools within five miles. As a result,  $\alpha_z$ , which I allow to be heterogeneous across household types, governs the return to effectiveness at raising student achievement and embeds incentives to locate in areas underserved by public schools. This heterogeneity may capture differences in preferences for school quality or reflect differences in information about the quality of schools across households. For potential entrants, whose effectiveness is realized post-entry, this heterogeneity also generates *ex ante* uncertainty.

The major identification concern in estimating equation (14) is endogeneity of the local market structure, including the number of nearby public and charter schools. Specifically, if entry and location choices are a function of unobserved demand determinants, absorbed into  $\xi_{it}^z$ , this classic simultaneity of demand and supply will lead to bias in the parameter estimates. For example, characteristics of locations, such as the non-test score quality, safety, or climate of local public schools, are observed by charter schools deciding where to locate, but remain unobserved to the econometrician. The resulting agglomeration of schools in unobservably attractive locations is likely to attenuate estimates of  $\gamma$  toward zero (or even make them positive), underestimating competitive impacts.

I assume that the unobservables in equation (14),  $\xi_{it}^z$ , can be decomposed into the sum of type-specific trends, unobserved location-specific characteristics, and an error term,

$$(15) \quad \xi_{it}^z = \tilde{\xi}_t^z + \tilde{\xi}_{j(i)}^z + \tilde{\xi}_{it}^z,$$

where the unobserved location-specific characteristics,  $\tilde{\xi}_{j(i)}^z$ , are fixed over time.<sup>35</sup> The key identifying assumption I make is that, conditional on the location fixed effects, unobserved demand determinants evolve independently of the presence and characteristics of nearby schools:

$$(16) \quad E\left[\tilde{\xi}_{it}^z \mid \mu_{it}, \bar{\mu}_{j(i)t}^P, NC_{j(i)t}^d, NP_{j(i)t}^d, x_{j(i)t}^D, \tilde{\xi}_t^z, \tilde{\xi}_{j(i)}^z\right] = 0.$$

Under this assumption, the local competitive environment is predetermined and thereby uncorrelated with the contemporaneous demand unobservables. This assumption is valid if schools' entry and exit choices are a function of the location fixed effects,  $\tilde{\xi}_{j(i)}^z$ , but not of the realization of the transitory shocks,  $\tilde{\xi}_{it}^z$ . Significant sunk costs in this setting, requiring for example that a charter school gain regulatory approval, secure facilities, hire staff, etc., in order to enter the market, suggest that this assumption is reasonable.<sup>36</sup>

<sup>35</sup>This decomposition presumes that the systematic unobserved panel variation in  $\xi_{it}^z$  is specific to a charter school's location rather than to the school itself. In other words, enrollment depends on only the observed school characteristics, chiefly effectiveness, in equation (14). Consistent with this, I find that estimates of the demand effects of school age and charter school type, e.g., No Excuses status, are not significantly different from zero.

<sup>36</sup>This assumption parallels the familiar restriction in empirical industrial organization that unobservable quality is (conditionally) independent of market structure, which implies that non-price attributes of competitors that may likely be fixed or costly-to-adjust in the short run are valid instruments for identifying substitution patterns (e.g., Berry, Levinsohn, and Pakes 1995). Gandhi and Houde (2016) contains a useful discussion. Houde (2012), for

Identifying assumption (16) suggests estimating demand by location-by-household type fixed effects. For parsimony, however, I bin each Census tract into one of ten types based on density and household income. I then estimate equation (14) by income quintile-by-urban-by-household type and school district-by-household type fixed effects. This assumes that the unobserved location fixed effects are constant across tracts of the same type in the same school district, a restriction that I test for robustness.<sup>37</sup> Intuitively, identification of the competitive effects relies on within-location type variation in enrollments and the local competitive environment. Within-location, competitive variation is driven in the model by idiosyncratic variation in schools' preferences and costs, including charters' private information draws.

*Variable Costs.*—To estimate the variable costs of charter school operation, I combine the estimates of charter school effectiveness with the panel of expenditures, student compositions, and location characteristics.

The specification I estimate is given by

$$(17) \quad \log VC_{it} = \kappa T(\log D_{it}, \mu_{it}) + \lambda \mathbf{Z}_{it} + \eta x_{j(i)t}^C + \epsilon_{it}.$$

The dependent variable in this equation is the reported total expenditure of charter school  $i$  during school year  $t$ , which depends on total enrollment, effectiveness, student composition  $\mathbf{Z}_{it}$ , and characteristics of the location,  $x_{j(i)t}^C$ .<sup>38</sup> The location characteristics I include in the estimation include local demographics as well as income quintile-urban and district-year intercepts. To allow for possible nonlinearities, such as a quantity-quality trade-off, I specify  $T(\log D_{it}, \mu_{it})$  by a polynomial in log enrollment and effectiveness. I also include indicators of charter school type to allow for heterogeneity in cost structure across charter schools. I estimate the cost function only on incumbent charter schools.

Cost estimation in education settings is subject to a number of endogeneity concerns (Costrell, Hanushek, and Loeb 2008; Duncombe and Yinger 2011; Gronberg, Jansen, and Taylor 2011). For example, effectiveness or quality of education is often poorly measured due to nonrandom sorting of students across schools. By estimating the education production (13) in a first stage, however, I obtain a measure of effectiveness identified by residual student achievement growth. As a result, the cost estimates obtained from equation (17) implicitly condition on students' unobserved prior inputs into education.<sup>39</sup>

Nonetheless, unobserved differences across schools in allocative efficiency may confound naïve regressions of expenditure on measures of outcomes and student

example, similarly uses variation in the number of and distance to local competitors in market-level data to identify spatial differentiation.

<sup>37</sup> I test this restriction by inclusion instead of Census tract fixed effects, which nonparametrically allow each location its own unobservable. Reassuringly, these results show little change in the demand estimates.

<sup>38</sup> In addition to the student compositions endogenized by the model (demographics and subsidized lunch status), I also control for the share of gifted, special education, and English language learner students in the cost estimation.

<sup>39</sup> Another traditional endogeneity concern is market power of public school districts in setting teacher wages. As charter schools are small players in the labor market, this is not a salient issue in this setting.

characteristics. In this case, the error term in (17) can be decomposed into an efficiency term and measurement error,

$$\epsilon_{it} = u_{it} + \zeta_{it}.$$

If efficiency in input usage, represented by  $u_{it}$ , is correlated with effectiveness or enrollment, then ordinary least squares estimates will be subject to omitted variable bias.<sup>40</sup> Teacher salary schedules, bureaucracy, and rent-seeking are likely sources of allocative inefficiency in public school districts. While the more competitive charter school market may lessen some of this worry, I also assume that any allocative inefficiency across charter schools is fully captured by the type of the charter school and its age, which I include in estimating equation (17).

### B. Entry Game

Estimating how charter schools respond to incentives relies on pairing the offline estimates of demand and variable costs with the logic of revealed preference in the entry game. This requires placing a functional form on the latent utility function of charter schools and a distributional assumption on the choice shocks. In this subsection, I also discuss the sources of identifying variation of the entry game parameters in the presence of unobserved market heterogeneity. Given the computational burden required to solve the game, I implement a two-step approach for estimation, described below.

I specify charter schools' utility as quasi-linear in net income per pupil,

$$(18) \quad U_{ij}(s_t) = \theta_i \Pi_{ij}(s_t) / D_{ij}(s_t) + g_i(D_{ij}(s_t), \mathbf{Z}_{ij}(s_t), \mu_{it}).$$

Therefore,  $\theta_i$  represents the marginal utility of net income per pupil and governs how charter schools, which are heterogeneous in their objectives by type, respond to financial incentives. On the other hand,  $g_i$  summarizes how, whether for altruistic or other reasons, charter schools may behave as if they value enrollment, student demographics, and effectiveness. Like  $\theta_i$ ,  $g_i$  is subscripted by  $i$  to highlight its dependence on charter school type. Using the equation for profit, I rewrite (18) as follows:

$$(19) \quad \begin{aligned} U_{ij}(s_t) &= \theta_i (\tau_t D_{ij}(s_t) - VC_{ij}(s_t) - FC_{jt}) / D_{ij}(s_t) + g_i(D_{ij}(s_t), \mathbf{Z}_{ij}(s_t), \mu_{it}) \\ &= \theta_i (\tau_t - VC_{ij}(s_t) / D_{ij}(s_t)) + \tilde{g}_i(FC_{jt}, D_{ij}(s_t), \mathbf{Z}_{ij}(s_t), \mu_{it}). \end{aligned}$$

This expression is useful for considering the potential sources of variation in the data that identify  $\theta_i$ , the key parameter of interest for the counterfactuals.

<sup>40</sup>The cost estimation literature typically models unobserved  $u_{it}$  using parametric distributional assumptions. See Davis and Garcés (2009, chapter 3) and Greene (2008) for overviews. Gronberg, Jansen, and Taylor (2012) is an application to charter schools.

For instance, consider the difference in expected utility for a potential entrant  $i$  choosing between locations  $j$  and  $j'$  (ignoring the preference shocks):

$$\begin{aligned} E[U_{ij}(s_t)] - E[U_{ij'}(s_t)] &= \theta_i \left( \tau_t - E[VC_{ij}(s_t)/D_{ij}(s_t)] \right) + E[\tilde{g}_i(FC_{jt}, D_{ij}(s_t), \mathbf{Z}_{ij}(s_t), \mu_{it})] \\ &\quad - \theta_i \left( \tau_t - E[VC_{ij'}(s_t)/D_{ij'}(s_t)] \right) + E[\tilde{g}_i(FC_{j't}, D_{ij'}(s_t), \mathbf{Z}_{ij'}(s_t), \mu_{it})] \\ &= \theta_i \left( -E[VC_{ij}(s_t)/D_{ij}(s_t)] + E[VC_{ij'}(s_t)/D_{ij'}(s_t)] \right) \\ &\quad + E[\tilde{g}_i(FC_{jt}, D_{ij}(s_t), \mathbf{Z}_{ij}(s_t), \mu_{it})] - E[\tilde{g}_i(FC_{j't}, D_{ij'}(s_t), \mathbf{Z}_{ij'}(s_t), \mu_{it})]. \end{aligned}$$

The per-pupil funding rate,  $\tau_t$ , drops out of the comparison. As a result,  $\theta_i$  is only informed by the sensitivity of entrants' location choice to expected variable cost differences (all else held equal). Potentially, however, the per-pupil funding rate  $\tau_t$  can inform the identification of  $\theta_i$  from the decision of whether to enter or of an incumbent to exit. Consider the latter choice in expected utility terms:

$$\begin{aligned} (20) \quad E[U_{ij}(s_t)] - E[U_{i0}(s_t)] &= \theta_i \left( \tau_t - E[VC_{ij}(s_t)/D_{ij}(s_t)] \right) + E[\tilde{g}_i(FC_{jt}, D_{ij}(s_t), \mathbf{Z}_{ij}(s_t), \mu_{it})]. \end{aligned}$$

The expected utility of exiting, denoted as alternative 0, is normalized to zero, so the comparison resolves to evaluating whether the expected utility of remaining in operation in location  $j$  (plus the difference in choice shocks) is positive. As can be seen, the statutory funding rate,  $\tau_t$ , like expected costs per pupil, informs  $\theta_i$  by shifting the expected profits of continued operation.

Unobserved private information draws accompany these expected utility comparisons. For this reason, it is important to consider the presence of unobserved heterogeneity, which can create omitted variable bias and is a principal focus in the empirical games literature (Seim 2006). Market-level unobservables, which I denote by  $\xi_t$ , also limit the variation for identifying  $\theta_i$ . To see this, consider again the incumbent decision, but incorporating unobserved heterogeneity:

$$\begin{aligned} E[U_{ij}(s_t)] - E[U_{i0}(s_t)] &= \theta_i \left( \tau_t - E[VC_{ij}(s_t)/D_{ij}(s_t)] \right) + E[\tilde{g}_i(FC_{jt}, D_{ij}(s_t), \mathbf{Z}_{ij}(s_t), \mu_{it})] + \xi_t. \end{aligned}$$

The expression makes clear that  $\xi_t$  is collinear with the statutory funding rate,  $\tau_t$ . As a result, taking unobserved heterogeneity seriously absorbs this source of variation in the data, leaving only expected variable costs per pupil to identify how charters respond to financial incentives. This highlights the importance of the expenditure data used to estimate variable costs offline.

I assume that the private information draws in (9) and (10) are distributed i.i.d. type I extreme value. This assumption produces familiar closed-form expressions

for the probability of continuing for incumbents (which I superscript by  $I$ ) and for choosing a location for potential entrants (superscripted  $E$ ). These are given by

$$(21) \quad P_{ij}^I(s_t) = \frac{\exp(\tilde{u}_{ij}(s_t, \mathbf{P}_t) + \xi_t)}{1 + \exp(\tilde{u}_{ij}(s_t, \mathbf{P}_t) + \xi_t)},$$

$$(22) \quad P_{ij}^E(s_t) = \frac{\exp(-EC_t + \tilde{u}_{ij}(s_t, \mathbf{P}_t) + \xi_t)}{1 + \sum_j \exp(-EC_t + \tilde{u}_{ij}(s_t, \mathbf{P}_t) + \xi_t)},$$

where the expected utility expressions are written to reflect the equilibrium structure of the choice environment. These choice probability expressions form the basis for the likelihood. As I observe multiple school districts for each time period, I capture the unobserved market-level heterogeneity in the estimation by directly including district-by-year fixed effects. A limitation is that the i.i.d. assumption restricts correlation in choices across charter schools and over time to arise from correlation in the market fixed effects,  $\xi_t$ , or the observed state variables. The potential restrictiveness of this assumption, however, is considerably relaxed by the heterogeneity in the model. The type-specific utility function, in particular, allows for correlation in choices across charter schools of the same type, while the persistence of type and effectiveness lead charter schools' choices to be correlated over time.

I specify  $\tilde{g}_i$  in equation (19) as a function of expected total enrollment, expected student demographics, effectiveness, and, to track fixed cost differences across locations, urban and income quintile intercepts and demographic shares of the location in the market. This function varies flexibly the type of charter school.<sup>41</sup> Entry costs are similarly allowed to be heterogeneous by charter school type (as well as district and year specific).

For estimation, I adopt a two-step estimator based on insights from Hotz and Miller (1993) to deal with the equilibrium structure of the model.<sup>42</sup> The first step semi-parametrically recovers conditional choice probabilities of entry location and exit,  $\hat{\mathbf{P}}_t$ . The second step then uses the choice probabilities as estimates of charter schools' beliefs to compute  $\tilde{u}_{ij}(s_t, \hat{\mathbf{P}}_t)$  offline using simulation. This two-step approach conditions estimation on the equilibrium played in the data.

I estimate the model by pooling the 26 school districts in Florida with at least 3 total charter school entries during the sample. A potential entrant's choice set consists of all Census tracts in their assigned school district.<sup>43</sup> As the demand model (6) overdetermines total enrollment, I use the predicted enrollment of the five mutually exclusive demographic groups (Asian, black, Hispanic, other, white) to predict

<sup>41</sup> I restrict the dependence of utility on effectiveness, via a quadratic polynomial, to be the same across the three types of charter schools, however. As these parameters primarily shift the exit probability, this choice is dictated by limited observed exits of for-profit and No Excuses charters in the data.

<sup>42</sup> Due to the large number of locations (e.g., there are nearly 350 Census tracts in Miami-Dade) and need to solve the game for all equilibria, a nested fixed point approach, adapted from Rust (1987), is computationally prohibitive. Two-step estimators have been extended to the estimation of dynamic games (Aguirregabiria and Mira 2007; Bajari, Benkard, and Levin 2007; Pakes, Ostrovsky, and Berry 2007). Bajari et al. (2010) apply the two-step estimator to a static game.

<sup>43</sup> I populate the pool of potential entrants for each school district by setting the total to equal the number of observed entrants plus the minimum of three-quarters the number of observed entries (rounded up to the next integer) and 3. I assign one-third of the created potential entrants to each type.

total enrollment in each location and require predicted enrollment to be at least one while capping the subsidized lunch share at one. For the first step of the estimation, I estimate a multinomial logit with a type-specific flexible polynomial in the state variables that includes district-by-year fixed effects. Finally, an entrant's initial effectiveness is drawn from the empirical distribution of  $\mu_1$ , which I discretize into a No Excuses and an other type distribution (with the restriction that the distributions share a common range).

## V. Results

This section presents the estimates of the empirical model and results of the counterfactual exercises. In the first subsection, I present the parameter estimates of the offline functions and from the entry game. As successive estimation steps use estimates obtained from prior ones, I compute standard errors using a block bootstrap that samples markets (district-years) with replacement. To assess goodness-of-fit, I then simulate the empirical model. I examine four counterfactual policy changes in the last subsection: (i) a cost-adjusted funding formula based on the estimated cost differentials; (ii) start-up grants targeted by household income of the location; (iii) a general increase in the per-pupil funding rate for charter schools; and (iv) a policy that restricts charter entry only to underserved locations. I compare these simulations in terms of aggregate effectiveness and the demographic composition of students served by the charter sector.

### A. Estimates

*Effectiveness.*—Figure 1 summarizes the school effectiveness estimates recovered from equation (13) by comparing public and charter schools operating in Florida in the 2012–2013 school year. I demean the effectiveness estimates over the entire sample to facilitate the comparisons and plot kernel densities for both school types. The figure illustrates that the average charter school was around 0.04 standard deviations (on the student achievement distribution) worse than average, with public schools ever so slightly better than average. Notably, the difference of the averages compares favorably with recent estimates of the relative quality of charter schools in Florida based on student-level data. For instance, CREDO (2009) reports charter school effects in the range of  $-0.03$  to  $-0.02$  standard deviations using matching. Similarly, Davis and Raymond (2012) obtain estimates in the range of  $-0.04$  to  $-0.02$  standard deviations from student fixed effects models. Figure 1 also reveals a wider dispersion in estimated effectiveness across charter schools than across public schools. These distributions are summarized numerically in the upper panel of Table 4. The estimates also reveal that No Excuses charter schools are more effective on average than other types of charters at raising student achievement. As summarized in Appendix Table A1, No Excuses charters are estimated to be  $0.14\sigma$  more effective on average. This magnitude is comparable with estimates of this difference from Massachusetts and New York City (Angrist, Pathak, and Walters 2013; Dobbie and Fryer 2013).

In Figure 2, I compare charter schools that exit the sample with charter schools that persist in terms of effectiveness. In contrast with the level test score difference

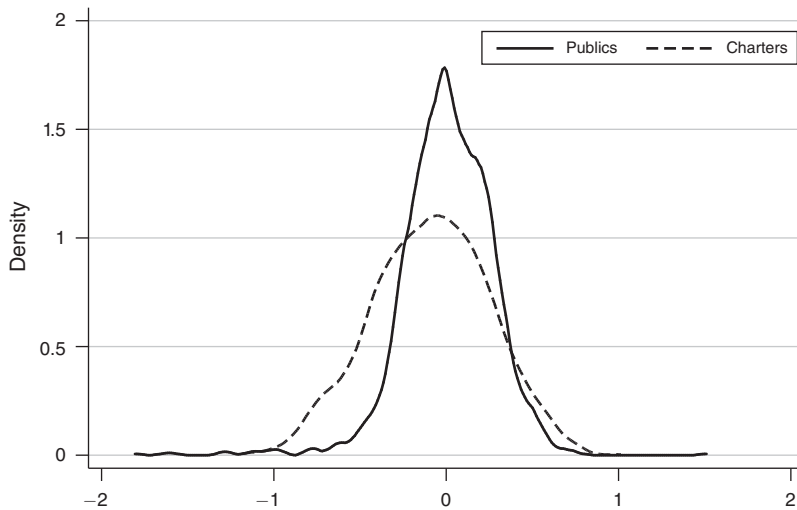


FIGURE 1. EFFECTIVENESS OF PUBLIC AND CHARTER SCHOOLS, 2012–2013

Note: Kernel densities of estimated effectivenesses of public and charter schools operating during the 2012–2013 school year.

TABLE 4—ESTIMATES OF SCHOOL EFFECTIVENESS

|                                         | Mean  | SD   | p5    | p25   | Median | p75   | p95  |
|-----------------------------------------|-------|------|-------|-------|--------|-------|------|
| <i>All schools, 2012–2013</i>           |       |      |       |       |        |       |      |
| Publics                                 | 0.01  | 0.26 | −0.36 | −0.14 | 0.01   | 0.18  | 0.40 |
| Charters                                | −0.04 | 0.34 | −0.61 | −0.27 | −0.03  | 0.19  | 0.51 |
| <i>Charter schools, 2006–2013</i>       |       |      |       |       |        |       |      |
| Persisters                              | 0.00  | 0.34 | −0.51 | −0.23 | 0.00   | 0.22  | 0.56 |
| Exiters                                 | −0.35 | 0.60 | −0.99 | −0.57 | −0.35  | −0.01 | 0.43 |
| <i>Nearby public schools, 2006–2013</i> |       |      |       |       |        |       |      |
| Persisters                              | 0.04  | 0.13 | −0.18 | −0.04 | 0.05   | 0.14  | 0.23 |
| Exiters                                 | 0.01  | 0.10 | −0.18 | −0.04 | 0.00   | 0.08  | 0.17 |

Notes: The top panel represents just 2012–2013 school year. In the middle panel, I use the shrunken fixed effect estimates of  $\mu_{1i}$  from (13) and predict effectiveness if each school were the average age of charter schools in 2012–2013. In the bottom panel, mean effectiveness is calculated over all public schools within 5 miles of each charter and averaged over the sample.

of nearly 1 standard deviation on average (summarized in Table 2), the density plots reveal a considerably smaller average gap in effectiveness at raising student achievement. As reported in the middle panel of Table 4, the average exiter charter school raises learning 0.35 standard deviations less than the average school. Moreover, the figure also shows considerable overlap in the effectiveness distributions, such that some exiter charter schools are very effective in absolute terms. For instance, the seventy-fifth percentile charter school that closes and exits the sample would be more effective than the average charter school operating in 2012–2013 if it survived. Conversely, the twenty-fifth percentile charter school that survives actually lowers student performance by about 0.23 standard deviations relative to the average school.

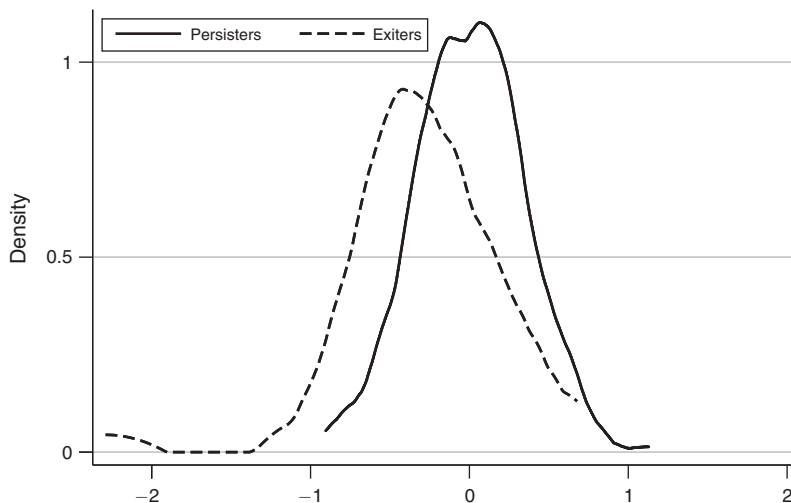


FIGURE 2. EFFECTIVENESS OF EXITERS AND PERSISTERS, 2006–2013

*Note:* Kernel densities of estimated effectivenesses of charter schools that exit and persist during the sample (fixing age at the 2012–2013 average).

The bottom panel of Table 4 presents a final comparison of persisting and exiting charter schools. Using the estimates of public school effectiveness recovered from equation (13), I compute the average effectiveness of public schools within five miles of each charter school and take the average over the sample period. This comparison produces the additional finding that exiters are located in relatively more underserved areas as measured by the educational quality of nearby public schools. Together, the estimated effectiveness distributions provide suggestive evidence that funding policies which sustain charter schools that would otherwise exit may not necessarily imply “costs” to the aggregate quality of charter schooling.

*Primitives.*—Demand estimates are presented in Tables 5 and 6. Estimates of competitive effects, shown in Table 5, indicate that charter school competition is highly localized. For instance, an additional competitor charter school within one mile is associated with an 11 percent drop in enrollment, but at further distances the competitive effect of other charter schools is not statistically different from zero. On the other hand, an additional public school within one mile is associated with nearly an 11 percent drop in enrollment, publics between one and three miles reduce enrollment around 4 percent, and publics three to five miles away reduce enrollment by 2 percent. These negative effects of competition that attenuate with distance provide a validity check of the estimates against economic theory.

Table 6 presents the estimates of the return to effectiveness and how demand depends on market size in equation (14). As expected, the estimates indicate that greater relative effectiveness is associated with larger charter school enrollment for most household types. For example, a one standard deviation increase in charter school effectiveness, all things being equal, is associated with around a 110 percent increase in the enrollment of Hispanic students. For black households, on the other

TABLE 5—ESTIMATES OF DEMAND FUNCTION: COMPETITIVE EFFECTS

|                                                 | log enrollment    |
|-------------------------------------------------|-------------------|
| Number of charters < 1 mile                     | −0.110<br>(0.039) |
| 1 mile ≤ number of charters < 3 miles           | 0.013<br>(0.017)  |
| 3 miles ≤ number of charters < 5 miles          | −0.000<br>(0.018) |
| Number of publics < 1 mile                      | −0.105<br>(0.030) |
| 1 mile ≤ number of publics < 3 miles            | −0.041<br>(0.007) |
| 3 miles ≤ number of publics < 5 miles           | −0.021<br>(0.005) |
| Income quintile × urban × HH type fixed effects | Y                 |
| District × HH type fixed effects                | Y                 |
| Year × HH type fixed effects                    | Y                 |
| Observations                                    | 8,590             |
| $R^2$                                           | 0.766             |

*Notes:* Standard errors obtained by block bootstrap that samples markets (100 draws). Specification also controls for relative effectiveness, log school-aged populations by household demographic in the 1-, 3-, and 5-mile distance bands, and dummy variables for year of entry and exit. Estimates of return to effectiveness, market size, and spillovers presented in Table 6.

hand, the estimate is not statistically different from zero. As relative effectiveness is with respect to local public schools (within five miles), these estimates influence charter schools' incentives to locate in underserved areas. Table 5 additionally displays how demand depends on school-aged populations within one mile. For example, a 10 percent increase in the size of the local Hispanic population is predicted to raise Hispanic enrollment in a charter school by over 2 percent. Reflecting peer spillovers, a larger school-aged Asian population is predicted to raise the enrollment of not only Asians, but also Hispanic and white students. Conversely, a larger black population in a given location is associated with lower enrollment of students from other demographics. These sorting interactions thus influence the equilibrium concentration of students across charter schools.

Table 7 presents estimates of the charter school variable cost function. The results indicate close to constant returns to scale of charter school operation. For a charter school of average size and effectiveness, a 10 percent increase in enrollment is associated with about a 9.8 percent increase in costs, all things being equal. While, all else held constant, a marginal increase in effectiveness does not appreciably change the costs of an average sized charter school, for-profit managed charter schools are estimated to have a substantial cost advantage over types of charter schools. In particular, for-profits are around 7 percent less expensive to operate than No Excuses charter schools, holding everything else the same. This difference may jointly reflect efficiencies of for-profit management and production function heterogeneity stemming from No Excuses charters schools' education model.<sup>44</sup>

<sup>44</sup>For instance, No Excuses charters tend to have longer school days and school years (Angrist, Pathak, and Walters 2013).

TABLE 6—ESTIMATES OF DEMAND FUNCTION: EFFECTIVENESS AND MARKET SIZE

|                                                 | log enrollment    |                   |                   |                   |                   |
|-------------------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                                                 | Asian             | Black             | Hispanic          | White             | FRP lunch         |
| Relative effectiveness                          | 0.625<br>(0.174)  | −0.179<br>(0.183) | 0.744<br>(0.208)  | 0.522<br>(0.306)  | 0.294<br>(0.152)  |
| <i>Within 1 mile</i>                            |                   |                   |                   |                   |                   |
| log Asian population                            | 0.116<br>(0.036)  | −0.069<br>(0.036) | 0.150<br>(0.038)  | 0.122<br>(0.046)  | 0.033<br>(0.033)  |
| log black population                            | −0.068<br>(0.032) | 0.142<br>(0.032)  | −0.203<br>(0.032) | −0.263<br>(0.052) | −0.004<br>(0.034) |
| log Hispanic population                         | −0.126<br>(0.052) | −0.17<br>(0.089)  | 0.225<br>(0.063)  | −0.029<br>(0.078) | 0.035<br>(0.050)  |
| log white population                            | 0.022<br>(0.066)  | 0.018<br>(0.088)  | −0.067<br>(0.068) | 0.171<br>(0.078)  | −0.129<br>(0.047) |
| Income quintile × urban × HH type fixed effects |                   |                   | Y                 |                   |                   |
| District × HH type fixed effects                |                   |                   | Y                 |                   |                   |
| Year × HH type fixed effects                    |                   |                   | Y                 |                   |                   |
| Observations                                    |                   |                   | 8,590             |                   |                   |
| $R^2$                                           |                   |                   | 0.766             |                   |                   |

*Notes:* Standard errors obtained by block bootstrap that samples markets (100 draws). Relative effectiveness refers to charter school's effectiveness less the average effectiveness of public schools within 5 miles. Specification also controls for number of charter and public schools by distance band, log school-aged populations by household demographic in the 3- and 5-mile distance bands, and dummy variables for year of entry and exit. Estimates for enrollment of other demographic students not shown. Estimates of the competitive effects are presented in Table 5.

The results in Table 7 also indicate cost differentials across student populations. In particular, holding all else equal, a 10 percentage point increase in the share of black students is associated with about a 3 percent increase in variables costs. For the share of Hispanic students, a 10 percentage point increase is associated with around a 2 percent increase in costs, all else constant. As these characteristics proxy for differences in socioeconomic status, household wealth, and other sources of advantage, they are highly collinear with subsidized lunch status.<sup>45</sup> I use these estimated cost differences to implement the counterfactual cost-adjusted funding formula, as described later.

*Utility.*—The results from estimating the entry game are presented in Table 8. These parameters correspond to the utility function of charter schools in deciding whether and where to open or to close. From the negative parameter on variable costs per pupil, the estimates reveal that charter schools respond positively to net income per pupil in their location and exit choices. The results also reveal there is heterogeneity across charter school types in objectives. In particular, No Excuses charter schools are estimated to be considerably less sensitive to variable costs per pupil than other types. In fact, the results cannot statistically reject that No Excuses'

<sup>45</sup> Though not presented in Table 7, the estimates imply that costs increase 3 percent on average for a 10 percentage point increase in the share of special education students, a recognized category of student need in the state funding formula.

TABLE 7—ESTIMATES OF VARIABLE COST FUNCTION

|                                       | log expenditure   |
|---------------------------------------|-------------------|
| log enrollment                        | 0.887<br>(0.101)  |
| log enrollment <sup>2</sup>           | 0.009<br>(0.009)  |
| Effectiveness                         | −0.081<br>(0.231) |
| Effectiveness <sup>2</sup>            | 0.033<br>(0.182)  |
| Effectiveness <sup>3</sup>            | 0.318<br>(0.191)  |
| Effectiveness × log enrollment        | −0.005<br>(0.039) |
| Percent FRP lunch                     | −0.001<br>(0.000) |
| Percent black                         | 0.003<br>(0.000)  |
| Percent Hispanic                      | 0.002<br>(0.001)  |
| Percent Asian                         | −0.004<br>(0.004) |
| Percent other                         | 0.006<br>(0.002)  |
| For-profit                            | −0.049<br>(0.025) |
| No Excuses                            | 0.018<br>(0.023)  |
| log age + 1                           | 0.012<br>(0.015)  |
| Income quintile × urban fixed effects | Y                 |
| District × year fixed effects         | Y                 |
| Observations                          | 1,153             |
| R <sup>2</sup>                        | 0.945             |

*Notes:* Standard errors obtained by block bootstrap that samples markets (100 draws). Specification also controls for the percentages of ESE, ELL, and gifted students, grade levels offered, Census tract demographics, the percentage of enrollment above grade 5, and a dummy for exit.

preferences do not consider net income in their decisions. On the other hand, revealed preference indicates that for-profit managed charter schools are the most income-oriented type.

Table 8 also presents estimates of how charters value expected enrollments and entry costs for 2012–2013. The utility estimates for No Excuses charter schools indicate preferences for smaller expected size and for serving black and Hispanic student populations, whereas these variables do not appear to influence other types of charter schools' choices. To facilitate interpretation, the point estimates can be put in willingness-to-pay terms. The results imply that a No Excuses charter school would be willing to pay around \$200 per pupil for a 1 percentage point increase in the share of black students in the school. In addition, the estimates indicate that the disutility of setting up a new (non-No Excuses and nonprofit) charter school

TABLE 8—ESTIMATES OF UTILITY FUNCTION

|                               | Utility            |                    |                   |
|-------------------------------|--------------------|--------------------|-------------------|
|                               | No Excuses         | For-profit         | Other             |
| Variable costs/pupil          | −1.401<br>(1.414)  | −3.479<br>(1.729)  | −2.297<br>(1.236) |
| Enrollment                    | −0.274<br>(0.130)  | 0.039<br>(0.059)   | 0.019<br>(0.048)  |
| Percent FRP lunch             | −0.007<br>(0.011)  | 0.001<br>(0.006)   | 0.000<br>(0.006)  |
| Percent black                 | 0.028<br>(0.014)   | 0.017<br>(0.011)   | −0.001<br>(0.008) |
| Percent Hispanic              | 0.047<br>(0.024)   | −0.010<br>(0.017)  | 0.001<br>(0.020)  |
| Entrant                       | −11.373<br>(2.088) | −11.428<br>(1.228) | −9.135<br>(0.749) |
| Entrant × year fixed effects  |                    | Y                  |                   |
| District × year fixed effects |                    | Y                  |                   |
| Observations                  |                    | 450,149            |                   |

*Notes:* Standard errors obtained by block bootstrap that samples markets (100 draws). *Variable costs/pupil* are in tens of thousands of dollars. The excluded entrant-by-year fixed effect is 2012–2013. Specification also includes a quadratic in effectiveness, interactions of charter school type with expected Percent Asian, local demographic shares, and urban and income quintile intercepts.

is worth nearly \$40,000 per pupil (over \$10 million for the average-sized charter school), indicative that barriers to entry into the market are considerable in magnitude. Estimated entry costs are larger in magnitude for No Excuses and for-profit managed charter schools. The estimates of the entry-by-year fixed effects, not presented in the table, show entry costs declining slightly over time.

It is worth considering potential biases in the utility estimates from not treating charter schools as forward-looking in their choices.<sup>46</sup> In this setting, the static setup is likely to underestimate the disutility of variable costs per pupil if charters are forward-looking. This is because the static setup neglects the option value of exiting in the future and the principal variables influencing choices are mean-reverting, making charter schools appear less sensitive to current variable cost differences.<sup>47</sup> The direction of these biases implies that the effects of funding formula policy changes can likely be viewed as lower bounds.

### B. Goodness-of-Fit

To assess the ability of the empirical model to fit the data, I compare selected moments to those generated by simulating the model. To perform the simulations, I begin with the data that characterizes the state of the education market in Florida

<sup>46</sup>Bayer et al. (2016) contains an especially useful discussion of such biases.

<sup>47</sup>The intuition for this is that the option value and mean-reversion both imply that choice alternatives are more alike in dynamic terms than they are in current characteristics. In the Appendix, I use the state variables with the variable cost and demand function estimates to predict costs per pupil, total enrollment, and student composition across all Census tract-years (for an arbitrary charter school) and estimate regressions of the form  $X_t - X_{t-1} = \theta X_{t-1} + \epsilon_{t-1}$ . In each case  $\theta < 0$ , indicating mean-reversion.

TABLE 9—MODEL FIT, 2012–2013

|                    | Data   | Model  |
|--------------------|--------|--------|
| Number of charters | 276    | 279.52 |
| Enrollment         | 328.98 | 321.25 |
| Percent FRP lunch  | 50.21  | 54.29  |
| Percent black      | 28.06  | 28.70  |
| Percent Hispanic   | 32.64  | 32.86  |
| Percent Asian      | 2.07   | 1.94   |
| Effectiveness      | −0.06  | −0.09  |

*Notes:* 2012–2013 averages over 50 simulated paths. Values represent average charter school (except for number of charters).

in the very first year of the sample, the 2006–2007 school year. Using the estimated parameters, I then simulate a number of paths forward to compare the average model predictions for the 2012–2013 school year to the 2012–2013 data. Simulation requires solving the model (i.e., finding a fixed point), so this exercise simplifies that requirement somewhat by only finding an equilibrium for states along each path.<sup>48</sup> The moments I examine are those of most interest in the counterfactual exercises, including the equilibrium number of charter schools, the composition of charter school enrollment, and aggregate effectiveness. Note that for the model to match these moments, it must accurately match the spatial configuration of the charter sector over time in addition to fitting enrollments.

Table 9 presents the comparisons. A total of 276 elementary charter schools with an average enrollment of 329 students were in operation in Florida in the 2012–2013 school year for the estimation sample. Matching these moment quite closely, the model predicts an average of 280 charter schools with an average enrollment of 321 students across the 50 simulated paths. Further, the predicted composition of students attending charter schools is close to the data: 50 percent of students in elementary charter schools participated in subsidized lunch, 28 percent were black and 33 percent were Hispanic in 2012–2013. The model predicts 54 percent, 29 percent, and 33 percent on average, respectively. Finally, aggregate charter school effectiveness, measured by the average across all charter schools in operation, is −0.06 in the data, as compared with 0.09 standard deviations below the mean predicted by the model. In sum, although the model underpredicts the aggregate effectiveness of the charter sector, these comparisons indicate that the model is able to reproduce important features of the data well.

### *C. Policy Counterfactuals*

In this section, I study counterfactual changes to funding policy in terms of equilibrium sector outcomes. I compare the counterfactuals in terms of two primary outcomes: the composition of students attending charter schools, determined by demand and where charter schools decide to locate, and the aggregate effectiveness

<sup>48</sup> Although the equilibrium found is not guaranteed to be unique, I initialize the contraction mapping using the first step conditional choice probability estimates, which represent the equilibrium played in the data, so the counterfactual equilibrium arrived at is “local.”

of the charter sector. By influencing entry and exit via competition, this outcome is sensitive to funding policy changes.

The first counterfactual implements a cost-adjusted funding formula designed to correct the strategic incentive to cream skim using the estimates of cost differences across students. These results thereby answer whether current funding policy has the distortionary effect of skewing the distribution of students served by charters toward low-cost student populations by influencing where charter schools locate. To implement this policy, I consider a funding formula that attaches weights to the enrollment of each household type and denote this policy by  $\hat{\tau}(\mathbf{D})$  where  $\mathbf{D}$  represents the vector of enrollments by household type. I choose the weights to satisfy two intuitive conditions. First, that the marginal revenue of enrolling an additional student of a given type approximately equals the marginal cost. This is where the variable cost estimates enter the calculation. The second condition is that the formula provides approximately the same amount of total revenue as the status quo funding formula:  $\hat{\tau}(\mathbf{D}) \approx \tau \mathbf{D}$ . In other words, the counterfactual is intended to only reallocate funding across household types. I use the average expenditure and student compositions in the data to calibrate the formula.<sup>49</sup>

I also consider two other counterfactual changes to funding policy. First, I study the effects of a targeted start-up grant that provides \$3,000 of additional support per pupil, on the order of existing federal and state grants, for charter schools locating in areas with average household income in the first quintile (below about \$45,000). This policy aims to capture the allocative benefits of the cost-adjusted funding formula in a practicable fashion. I then examine a general per-pupil funding increase to quantify the elasticities of charter school supply and aggregate effectiveness. This counterfactual is interesting given its two likely effects. On the one hand, the additional funding may intensify competition with additional entry, raising sector quality through differential exit. This is countervailed against the possibility that, with spatial competition, the additional funding subsidizes low-effectiveness schools.

I compare the equilibrium effects of the three funding policy counterfactuals described above with an additional policy that instead regulates where new charter schools may open. The policy implemented restricts charter entry only to locations in which the nearby public schools are low-performing (i.e., first or second average test score quintile). This counterfactual thus simulates the effects in Florida of an institutional environment that would sharply limit charter entry only to underserved neighborhoods.<sup>50</sup> As shown in Table 10, these neighborhoods are disproportionately urban, low income, and minority (particularly black). This counterfactual thus provides a valuable comparison for quantifying the effects of the funding formula changes on equilibrium outcomes, particularly charter student demographics.

Table 11 presents the results of the counterfactual simulations, which are implemented by the same procedure used to assess model fit. I report the effects in terms of percentage changes from the model predictions reported in Table 9. Column 1 reports the effects of a cost-adjusted funding formula. The cost-adjustment leads to a

<sup>49</sup>To provide a sense of the magnitudes, the difference in funding for a fully Hispanic charter school and a fully white charter school is about \$1,900 per pupil in the implemented counterfactual.

<sup>50</sup>This policy therefore mimics the effects of features of, for example, New York's institutional environment, which weighs community need in approving new charter schools.

TABLE 10—CHARACTERISTICS OF CENSUS TRACTS BY NEARBY PUBLIC SCHOOL PERFORMANCE

| Quintile | Density | HH income | Percent black | Percent Hispanic | Math $\bar{z}$ | Reading $\bar{z}$ | $\bar{\mu}$ |
|----------|---------|-----------|---------------|------------------|----------------|-------------------|-------------|
| 1        | 4,851   | 51,428    | 34.19         | 20.96            | -0.54          | -0.68             | -0.05       |
| 2        | 4,610   | 57,053    | 19.48         | 22.62            | -0.18          | -0.28             | -0.01       |
| 3        | 2,780   | 60,898    | 13.35         | 15.03            | -0.03          | -0.06             | -0.02       |
| 4        | 3,526   | 68,755    | 9.48          | 23.09            | 0.19           | 0.18              | 0.05        |
| 5        | 2,307   | 79,859    | 6.43          | 10.81            | 0.57           | 0.66              | 0.12        |

Notes: *Density*, *Household income*, and demographics represent area within one mile of the Census tract, math and reading scores represent average normed fourth and fifth grade exam performance for public schools within five miles, and  $\mu$  is the average estimated effectiveness of public schools within five miles. Quintiles are defined by 2006–2007 reading test scores.

TABLE 11—COUNTERFACTUAL RESULTS, 2012–2013

|                    | (1)   | (2)   | (3)   | (4)    |
|--------------------|-------|-------|-------|--------|
| Number of charters | 1.97  | 6.02  | 11.55 | -24.60 |
| Enrollment         | -2.79 | -0.35 | -0.26 | -9.56  |
| Percent FRP lunch  | 3.64  | 3.34  | -0.12 | 12.74  |
| Percent black      | 7.47  | 5.78  | 0.29  | 16.14  |
| Percent Hispanic   | 1.18  | 0.31  | 0.32  | -2.31  |
| Percent Asian      | -6.14 | 0.31  | 1.71  | -9.85  |
| Effectiveness      | -0.01 | -0.01 | 0.00  | 0.02   |

Notes: Average percentage change from model predictions over 50 simulated paths, 2012–2013 school year. Effectiveness measured by average across charter schools and reported as difference in level. Column 1 corresponds to a cost-adjusted funding formula using estimates of variable cost differences calibrated to leave average revenue unchanged; column 2 corresponds to start-up grant for locations with mean household income in the first quintile (less than approx. \$45,000) of \$3,000 per pupil; column 3 corresponds to additional \$1,000 per pupil in operating revenue for all charters; column 4 corresponds to a policy that restricts charter entry only to locations with first or second quintile local public school performance.

significant shift in the demographics of charter school students. In the counterfactual, the share of students attending charters that participate in subsidized lunch increases by around 4 percent and the share of charter students who are black increases by over 7 percent. These results provide evidence of cost-based cream skimming by charter schools. The cost-adjusted funding formula also generates a small decline in the aggregate effectiveness of the charter sector. As suggested earlier by Table 4, this is in large part due to the fact that a number of the charter schools sustained in the market by this policy change are not relatively ineffective.

To further get at the mechanisms behind these changes, Table 12 presents changes in the student demographics of entrants and exiters across the counterfactuals. These results highlight location choice as an important margin for charters to target low-cost student populations: the top panel, which summarizes the student demographics of all entrants, shows that, under the cost-adjusted funding policy (column 1), entering charter schools choose locations such that the average share of students they serve that participate in subsidized lunch increases by about 5 percent and the share who are black increases by over 8 percent. Moreover, the table reveals that cream-skimming is primarily driven by non-No Excuses charter schools. Consistent with the estimated differences in objectives, the location choices (and thus student demographics) of entering No Excuses charter schools are less responsive to the policy.

TABLE 12—ENTRANTS AND EXITERS IN COUNTERFACTUALS

|                            | (1)   | (2)   | (3)   | (4)    |
|----------------------------|-------|-------|-------|--------|
| <i>All entrants</i>        |       |       |       |        |
| Percent FRP lunch          | 4.62  | 6.25  | −0.29 | 25.05  |
| Percent black              | 8.29  | 7.18  | −0.17 | 30.05  |
| Percent Hispanic           | 1.52  | −0.60 | −0.05 | −1.83  |
| Percent Asian              | −9.58 | −7.08 | −3.00 | −17.97 |
| <i>No Excuses entrants</i> |       |       |       |        |
| Percent FRP lunch          | 2.20  | 1.10  | −1.16 | 22.86  |
| Percent black              | 4.76  | 2.34  | −1.75 | 22.72  |
| Percent Hispanic           | 2.41  | 1.21  | 0.31  | −1.50  |
| Percent Asian              | −2.62 | −3.18 | 0.71  | −14.11 |
| <i>All exiters</i>         |       |       |       |        |
| Percent FRP lunch          | −2.10 | 1.97  | 0.07  | 4.66   |
| Percent black              | −4.46 | 1.36  | 1.19  | 6.01   |
| Percent Hispanic           | −1.74 | −0.26 | −2.71 | −3.97  |
| Percent Asian              | −0.30 | 3.10  | −1.96 | −5.64  |

*Notes:* Average percentage change from model predictions over 50 simulated paths. Column 1 corresponds to a cost-adjusted funding formula using estimates of variable cost differences calibrated to leave average revenue unchanged; column 2 corresponds to start-up grant for locations with mean household income in the first quintile (less than approx. \$45,000) of \$3,000 per pupil; column 3 corresponds to additional \$1,000 per pupil in operating revenue for all charters; column 4 corresponds to a policy that restricts charter entry only to locations with first or second quintile local public school performance.

The second counterfactual I study, presented in column 2, is a targeted start-up grant for charter schools that locate in areas with average household income in the first quintile. As shown in Table 11, this policy change yields a comparable change in the demographics of charter students as the cost-adjusted funding formula (column 1) (e.g., the share of students attending charters who are black increases about 6 percent). The entry grant also expands the sector by 6 percent. Table 12 illustrates that the mechanism behind the demographic shift is somewhat different. While the student demographics of entrants' similarly shift under both policies, the bottom panel reveals that the subsidized lunch and black shares of charter schools that exit do not change or even increase under counterfactual (column 2). This is because, unlike the cost-adjustment, the start-up grant does not support charter schools as incumbents, which has the effect of raising the exit rate somewhat of charter schools reallocated by the policy.

The results of increasing the annual per-pupil funding rate by \$1,000 per pupil, the third funding counterfactual, are reported in column 3 of Table 11. This additional funding substantially increases the number of charter schools by almost 12 percent, which translates to around 32 additional charter schools in 2012–2013. Aggregate effectiveness of the charter sector, however, does not change appreciably with the additional entry. This suggests that the effect of subsidizing weak competitors dominates, as about 6 fewer charter schools exit in this counterfactual on average. These results reinforce the general finding of limited competitive incentives.

Finally, column 4 of Table 11 presents the results of a policy that, instead of manipulating the funding formula, restricts charter entry only to locations in which the nearby public schools are low-performing. The table reveals that this policy leads to a large shift in the demographics of charter school students: the policy

raises the share of charter students that participate in subsidized lunch by 13 percent and the share of charter students who are black by 16 percent. As a result, the cost-adjusted funding formula closes about 30–50 percent of the discrepancy in charter student demographics with a policy that restricts charters only to underserved neighborhoods. This highlights the significant inequities that are generated by the funding distortion.<sup>51</sup>

## VI. Conclusion

This paper uses unique data from Florida to study the effects of charter school funding policy on the equilibrium quality and demographics of the charter sector. This is motivated by the possibility that, due to the flat statutory funding formula, incentives embedded in charter schools' choices of location or to exit skew the distribution of students served by charters toward low-cost student populations. The data assembled present suggestive differences in this regard: exiters in the sample serve 21 percentage point higher subsidized lunch and 37 percentage point higher black student bodies on average. Although they display much lower student achievement on average, charter schools that exit also spend over \$900 more per student per year.

To understand the mechanisms behind these patterns, I develop and estimate an empirical model of charter school supply and competition that enables me to study the effects of counterfactual funding policies on equilibrium sector outcomes. In the model, charters consider competition and the cost of educating the expected student body composition in choosing where to locate. As a result, charter sector outcomes depend on whether and how charter schools strategically respond to the policy environment. The model incorporates No Excuses and for-profit charter school types that are heterogeneous in their cost structure and objectives. To estimate the model, I first recover estimates of school effectiveness, demand, and variables costs, which embed cost differences across student populations, to estimate how charter schools respond to financial and competitive incentives via revealed preference.

I use the estimated model to study three funding policy changes in detail. First, I consider a cost-adjusted funding formula that aims to correct the strategic incentive to cream skim. The results indicate a significant increase in the shares of subsidized lunch and black students attending charter schools with little reduction in aggregate effectiveness. This result provides evidence that the current flat funding formula unintentionally leads charter schools, in particular non-No Excuses charter schools, to underserve disadvantaged students. Compared with a policy that instead restricts new charter entry only to locations in which local public schools are low-performing, the cost-adjusted funding formula closes about 30–50 percent of the discrepancy in charter student demographics. I also consider a location targeted start-up grant that provides support for charter schools the locate in lower income areas, which yields a comparable shift in student demographics, and examine the effects of a general

<sup>51</sup> While the restricted entry counterfactual is used primarily to benchmark the funding formula effects, the policy also dramatically reduces the size of the charter sector. The implication is that, while a larger *share* of students attending charter schools under this policy are from disadvantaged backgrounds, fewer overall actually attend charter schools. As entrants have lower effectiveness than average, the restricted entry counterfactual also raises the aggregate effectiveness of the charter sector.

funding rate increase for charters. The latter policy increases the size of the sector, but generates little change in the aggregate effectiveness of the sector.

These findings are important as they are informative about school choice policy. In particular, a mismatch between funding and costs may generate significant disparities in access to and benefits from school choice. This point, while relevant to ongoing debates over further expansion of charter and voucher programs, is largely unrecognized in the existing literature on education markets. Further, the counterfactuals I study underscore that funding policy, an element common to both private school voucher and charter school programs, may provide an effective policy instrument for directing competition via supply-side incentives. This has implications for the design of school choice programs broadly and contributes to a growing empirical literature on subsidized private (or nonpublic) provision of education. For example, this paper focuses on the equilibrium implications of funding policy in Florida via the margin of school entry and exit, while Neilson (2013) argues that a modification of voucher funding rules in Chile led to improvements via the intensive margin of quality. The policy change created supply-side incentives through demand by raising the voucher amount for poor students (and eliminating fees for many others). The policy considered in this paper is targeted at funding and cost distortions that likewise limit access to quality schooling for underprivileged students. Along related lines, Abdulkadiroğlu, Pathak, and Walters (2018) present evidence of negative selection by private schools into Louisiana's voucher program, while Andrabi et al. (2017) study how private schools competitively respond to funding grants in rural Pakistan.

The empirical model I develop makes several simplifying assumptions that motivate future work. For instance, I largely abstract from how households choose schools. Although features such as information influence how students sort across schools, they also have supply-side implications that I do not develop here. In addition, I do not analyze the sources of within-school changes in effectiveness over time. While on the one hand this may underestimate quality effects of competition, it also suggests another margin which may inform the design of school choice policy, particularly school accountability programs. Lastly, as the charter sector continues to mature, the roles of charter school networks and management organizations, which are beyond the scope of this work, are likely to take on added importance.

## APPENDIX A

A1. *No Excuses Charter Schools*

TABLE A1—DENSITY AND EFFECTIVENESS OF NO EXCUSES CHARTER SCHOOLS, 2006–2013

|               | No Excuses      | Non-No Excuses  | Difference |
|---------------|-----------------|-----------------|------------|
| Density       | 5,330<br>(398)  | 3,828<br>(304)  | 1,501      |
| Effectiveness | −0.01<br>(0.04) | −0.16<br>(0.03) | 0.14       |

*Notes:* Robust standard errors in parentheses. Density corresponds to area within one mile of the school's Census tract.

I create an indicator variable for whether each Florida charter school is No Excuses using information about mission and instructional practices from a statutorily-mandated annual survey by the Florida Office of Independent Education and Parental Choice. Drawing upon the associations of school philosophy variables with No Excuses status for Boston charter schools reported in Angrist, Pathak, and Walters (2013), I classify a Florida charter operator as No Excuses if it follows a college preparatory curriculum, emphasizes core math and reading skills in its mission, or the school focuses on STEM and also emphasizes either discipline or achievement. I rule out charters that are Montessori, language immersion, arts focused, or report using blended instruction as No Excuses.

The indicator created matches key characteristics about No Excuses charter schools from the literature. As shown in Table A1, No Excuses charter schools in Florida are both substantially more urban (located in areas with around 1,500 more people per square mile on average) and are  $0.14\sigma$  more effective at raising student learning than other charters on average. Both differences are statistically significant. By comparison, Angrist, Pathak, and Walters (2013) reports that Massachusetts No Excuses charters are  $0.21\sigma$  and  $0.15\sigma$  more effective in math and reading, respectively. Similarly, Dobbie and Fryer (2013) report differences in the range of  $0.04$ – $0.08\sigma$  greater effectiveness for New York City charters that follow No Excuses practices.

## A2. Exit Dynamics

TABLE A2—CHARTER SCHOOL EXIT DYNAMICS

|                  | Enrollment        |                   | Math $z$           |                  | Reading $z$        |                   | Percent FRP lunch   |                    |
|------------------|-------------------|-------------------|--------------------|------------------|--------------------|-------------------|---------------------|--------------------|
|                  | (1)               | (2)               | (3)                | (4)              | (5)                | (6)               | (7)                 | (8)                |
| Years until exit | -15.77<br>(4.460) | -22.36<br>(24.87) | -0.245<br>(0.0654) | 0.129<br>(0.365) | -0.262<br>(0.0663) | 0.0925<br>(0.299) | 0.580<br>(2.005)    | -0.524<br>(4.632)  |
| Year FE          | Y                 | Y                 | Y                  | Y                | Y                  | Y                 | Y                   | Y                  |
| School FE        | N                 | Y                 | N                  | Y                | N                  | Y                 | N                   | Y                  |
|                  | Percent white     |                   | Percent black      |                  | Percent Hispanic   |                   | Percent Asian       |                    |
|                  | (9)               | (10)              | (11)               | (12)             | (13)               | (14)              | (15)                | (16)               |
| Years until exit | 0.113<br>(0.510)  | -3.223<br>(3.045) | -0.916<br>(0.620)  | 3.953<br>(4.124) | 1.089<br>(0.496)   | -0.750<br>(1.635) | -0.0756<br>(0.0612) | -0.0907<br>(0.323) |
| Year FE          | Y                 | Y                 | Y                  | Y                | Y                  | Y                 | Y                   | Y                  |
| School FE        | N                 | Y                 | N                  | Y                | N                  | Y                 | N                   | Y                  |

Notes: Robust standard errors in parentheses. Table examines exiter charter schools' characteristics leading up to exit from the sample. Even numbered specifications contain 93 charter school-year observations (which exclude the initial year each exiter appears in the sample) and control for charter school age, whether an entrant, and the initial value of the dependent variable. Odd numbered, school fixed effects specifications contain 160 observations and include an indicator for year of entry. Observations in (3) through (6) are 71, 109, 73, and 112, respectively.

## A3. Mean Reversion

TABLE A3—MEAN REVERSION OF PREDICTED SCHOOL CHARACTERISTICS

|              | $\Delta VC/pupil$  | $\Delta Enrollment$ | $\Delta \% FRP lunch$ | $\Delta \% Asian$  | $\Delta \% black$  | $\Delta \% Hispanic$ |
|--------------|--------------------|---------------------|-----------------------|--------------------|--------------------|----------------------|
|              | (1)                | (2)                 | (3)                   | (4)                | (5)                | (6)                  |
| Lagged level | -0.673<br>(0.0488) | -0.633<br>(0.0626)  | -0.672<br>(0.0169)    | -0.373<br>(0.0852) | -0.882<br>(0.0212) | -0.681<br>(0.0285)   |

Notes: Robust standard errors in parentheses. Table estimates regressions of the form  $X_t - X_{t-l} = \theta X_{t-1} + \epsilon_{t-1}$ , where the location state variables and demand and variable cost function estimates are used to predict school characteristics for an arbitrary charter school for all Census tract-years (absent competition from other charters). All columns include school district-by-year and Census tract fixed effects. Each column contains 15,828 observations.

## A4. Additional Summary Statistics

TABLE A4—LOCATION AND STUDENT SUMMARY STATISTICS FOR CHARTER SCHOOLS, 2012–2013

|                                 | Mean   | SD     | p10    | Median | p90    |
|---------------------------------|--------|--------|--------|--------|--------|
| <i>Location characteristics</i> |        |        |        |        |        |
| Density                         | 3,722  | 3,752  | 209    | 2,983  | 7,507  |
| Household income                | 59,206 | 21,536 | 37,554 | 53,836 | 89,680 |
| Percent white                   | 45.59  | 29.85  | 2.82   | 47.61  | 85.03  |
| Percent black                   | 20.84  | 23.58  | 0.60   | 11.75  | 54.97  |
| Percent Hispanic                | 29.84  | 27.21  | 3.67   | 20.27  | 72.30  |
| Percent Asian                   | 2.23   | 2.27   | 0.05   | 1.62   | 4.74   |
| <i>Student characteristics</i>  |        |        |        |        |        |
| Enrollment                      | 324.76 | 263.57 | 72     | 234    | 687    |
| Percent FRP lunch               | 50.05  | 27.16  | 13.64  | 49.77  | 87.52  |
| Percent white                   | 34.86  | 29.59  | 1.20   | 30.19  | 78.05  |
| Percent black                   | 27.49  | 30.73  | 1.06   | 13.26  | 87.76  |
| Percent Hispanic                | 31.91  | 29.03  | 3.85   | 21.50  | 84.38  |
| Percent Asian                   | 2.07   | 2.85   | 0      | 1.32   | 4.76   |
| <i>School characteristics</i>   |        |        |        |        |        |
| Expenditure/pupil               | 7,398  | 1,932  | 6,034  | 7,051  | 9,407  |
| Math z                          | −0.05  | 1.11   | −1.49  | 0.00   | 1.38   |
| Reading z                       | 0.14   | 1.05   | −1.30  | 0.18   | 1.36   |
| No Excuses                      | 0.18   | 0.38   | 0      | 0      | 1      |
| For-profit                      | 0.36   | 0.48   | 0      | 0      | 1      |

Notes: Observations are for 288 charter schools. Location characteristics represent area within one mile of a school's Census tract. Expenditure missing for 13 schools.

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